

Attention and Transformers

Prof. Jianguo Zhang
SUSTech

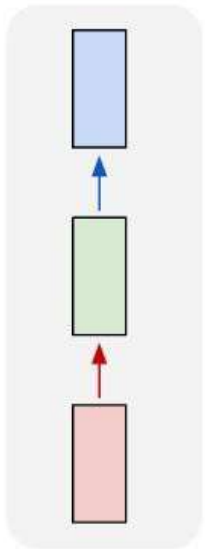
Slides were adapted from various sources

1. From RNN to Attention Layers

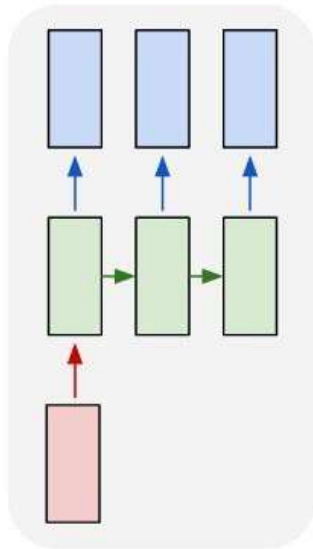
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Last Time: Recurrent Neural Networks

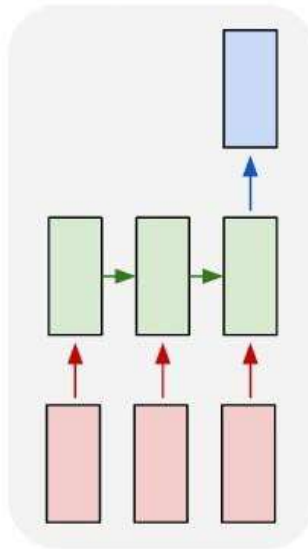
one to one



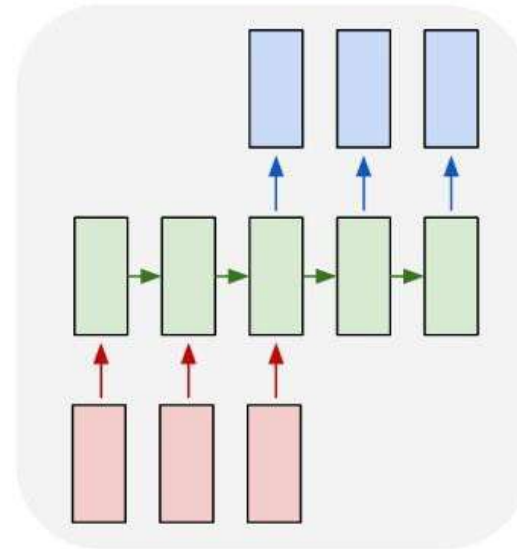
one to many



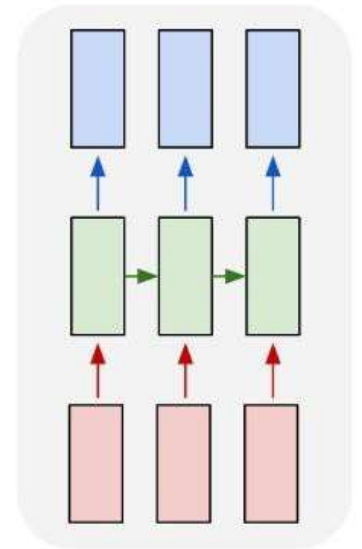
many to one



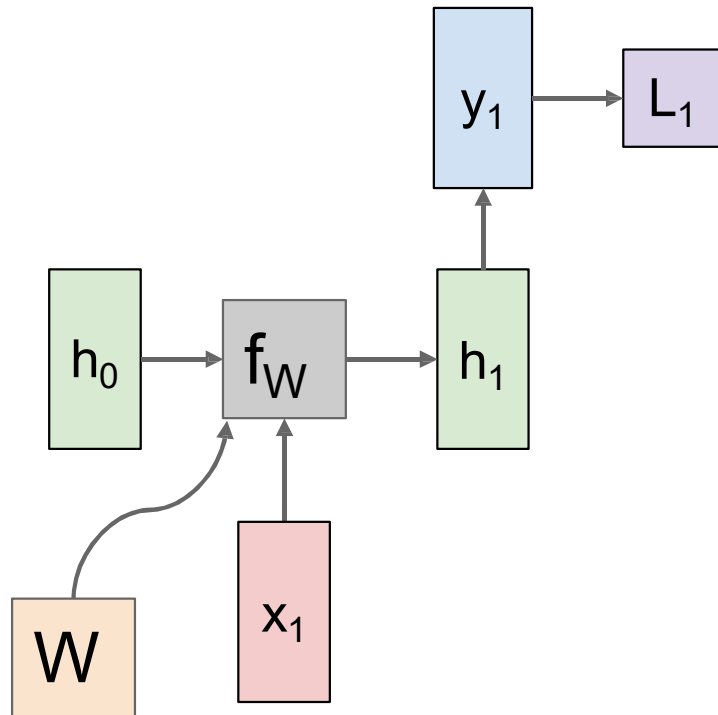
many to many



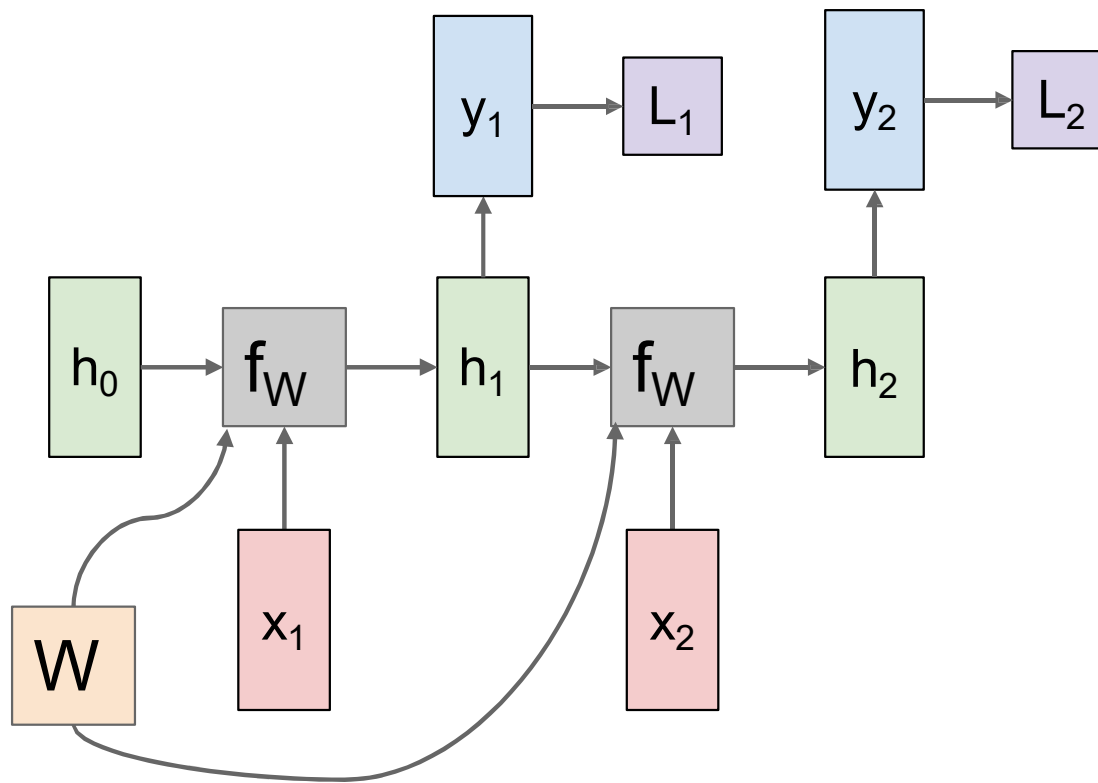
many to many



Last Time: Variable length computation graph with shared weights

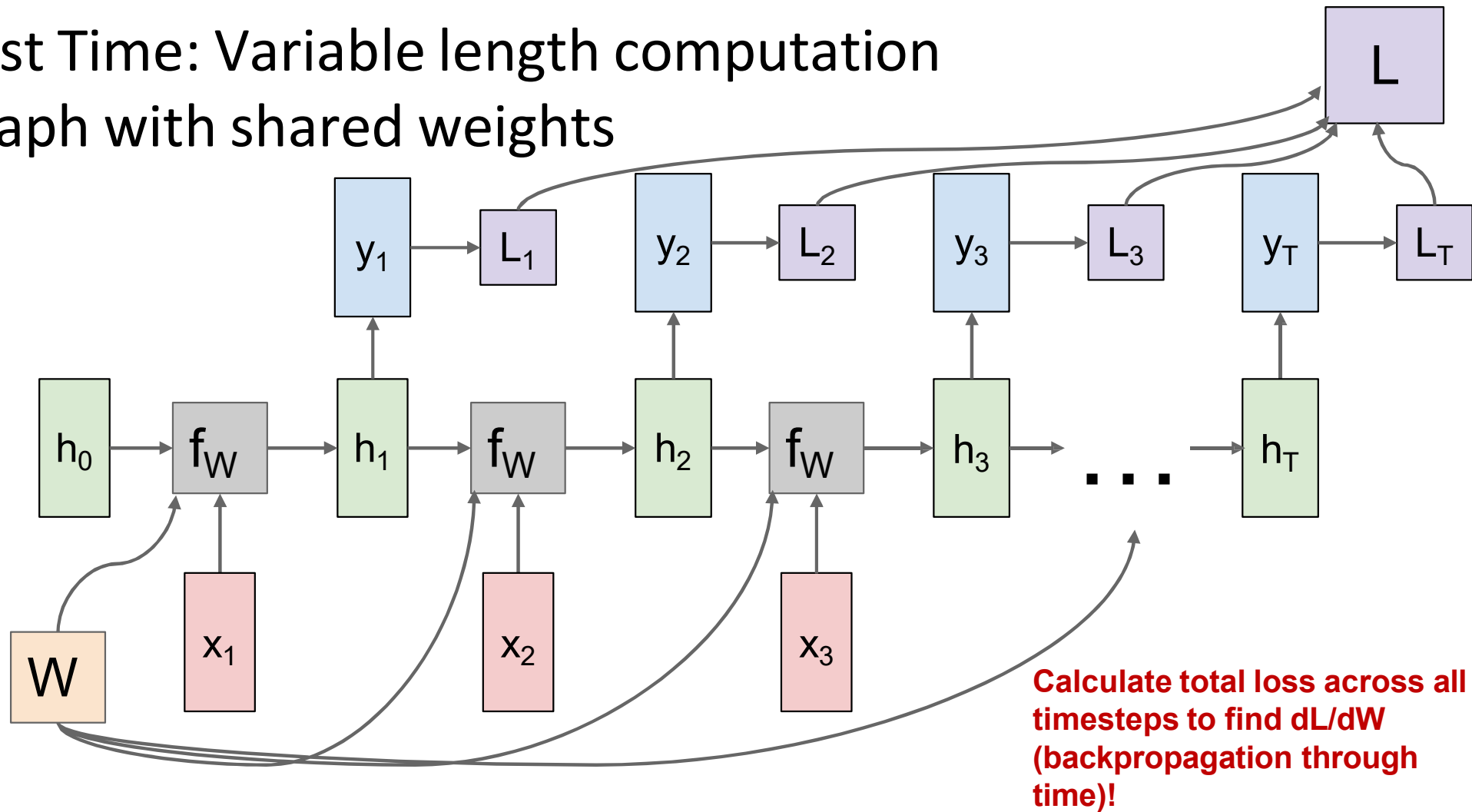


Last Time: Variable length computation graph with shared weights



W is reused (recurrently)!

Last Time: Variable length computation graph with shared weights



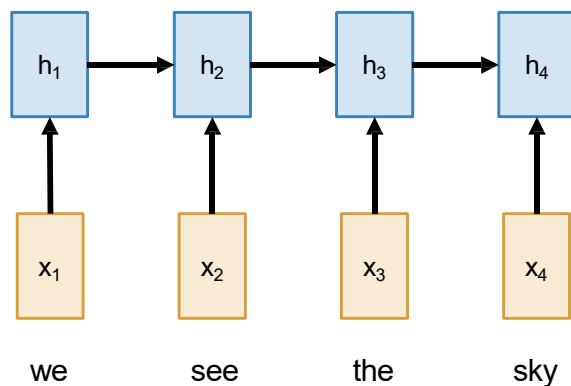
Sequence to Sequence with RNNs: Encoder - Decoder

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

A motivating example for today's discussion –
machine translation! English $\rightarrow$ Italian

Encoder: $h_t = f_W(x_t, h_{t-1})$



Sequence to Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

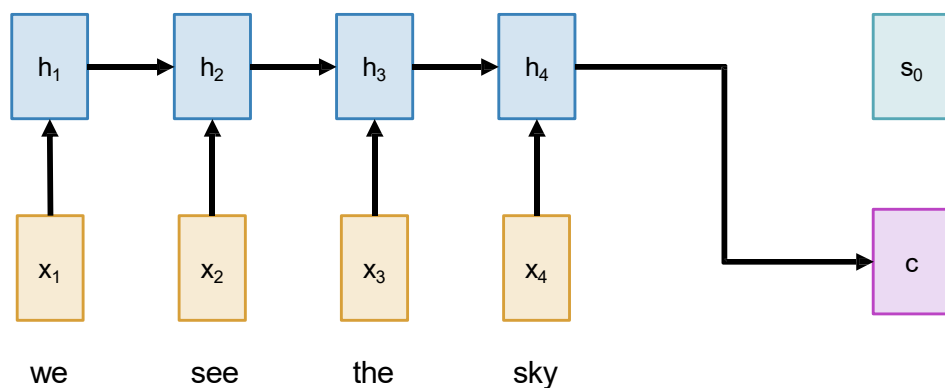
Output: Sequence $y_1, \dots, y_T$

From final hidden state predict:

Encoder: $h_t = f_W(x_t, h_{t-1})$

Initial decoder state s_0

Context vector c (often $c=h_T$)

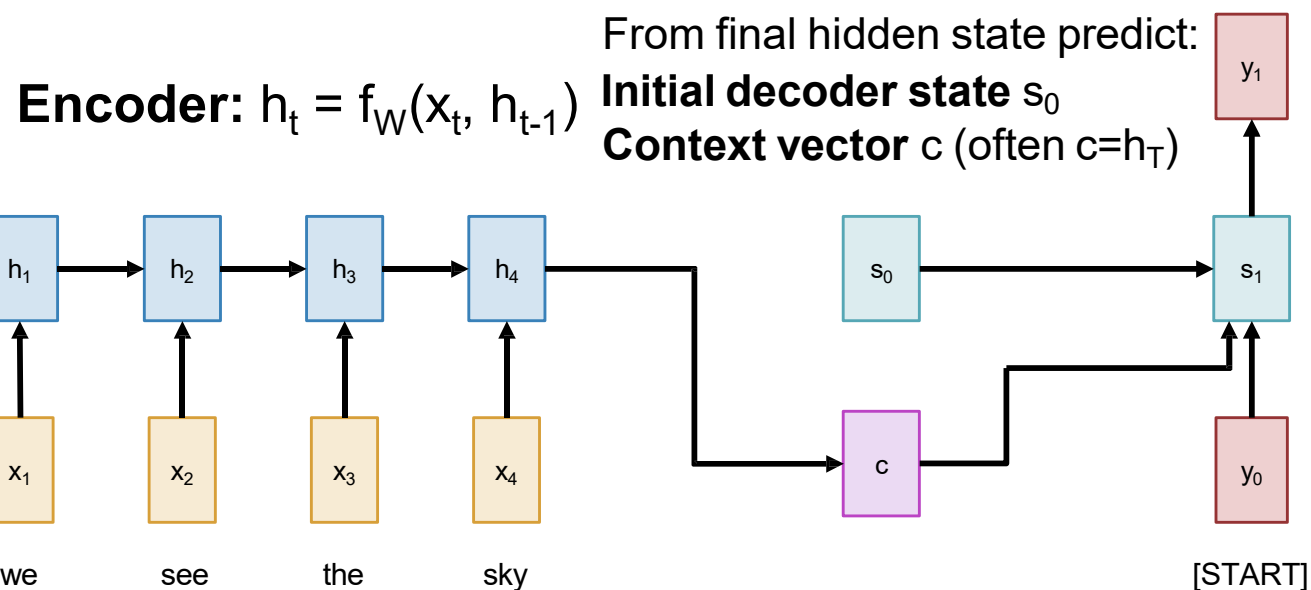


Sequence to Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$



Sequence to Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

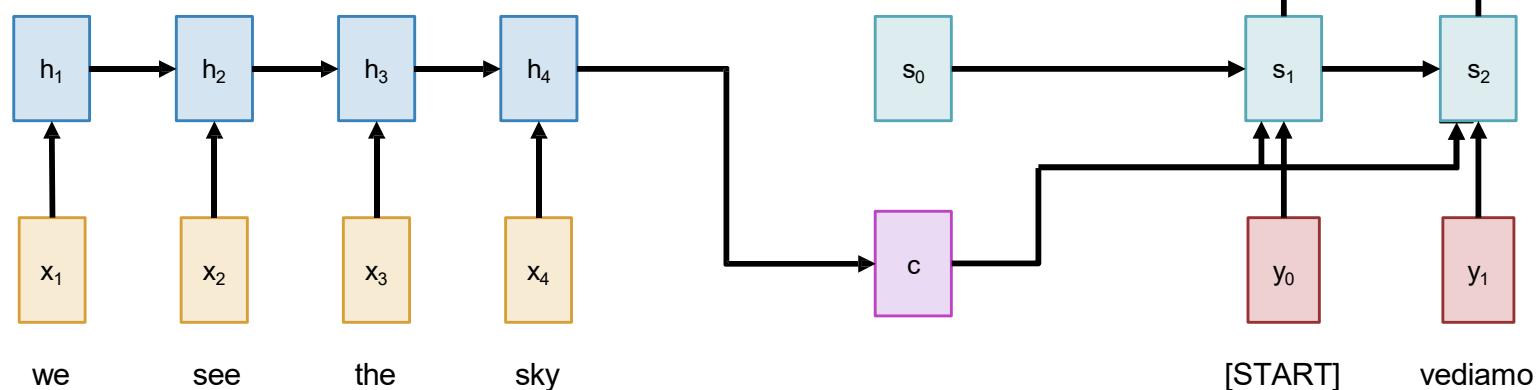
Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$

Encoder: $h_t = f_W(x_t, h_{t-1})$

From final hidden state predict:

Initial decoder state s_0

Context vector c (often $c=h_T$)



Sequence to Sequence with RNNs

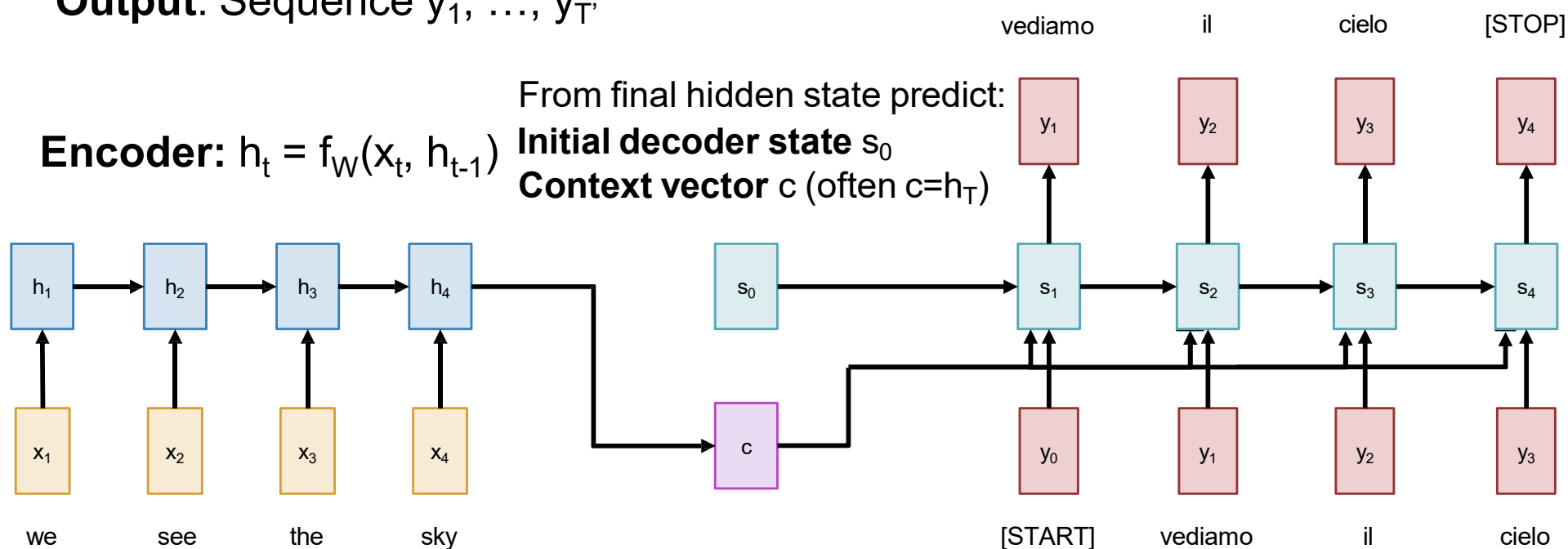
Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_{T'}$

Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$

Encoder: $h_t = f_W(x_t, h_{t-1})$

From final hidden state predict:
Initial decoder state s_0
Context vector c (often $c=h_T$)



Sequence to Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

Remember:

During Training:

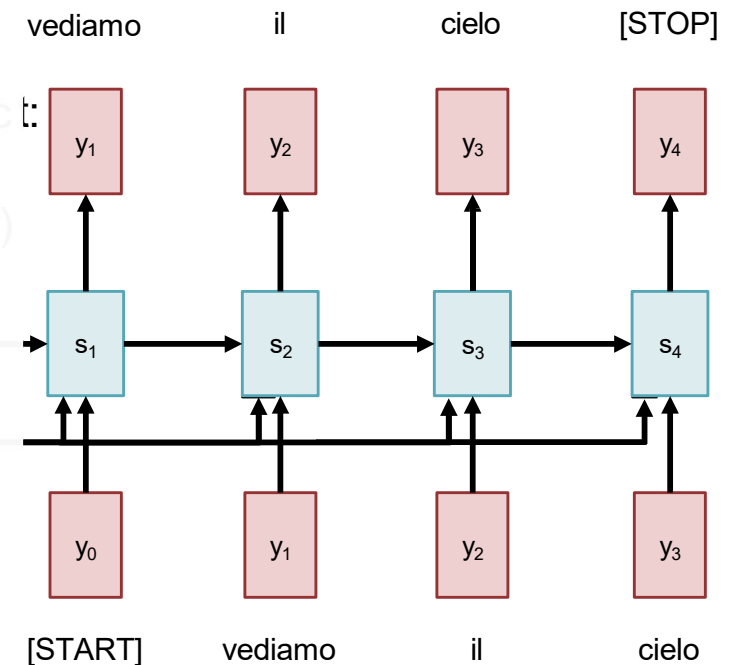
Often, we use the “correct” token even if the model is wrong. Called **teacher forcing**

During Test-time:

We sample from the model’s outputs until we sample [STOP]



Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$



Sequence to Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

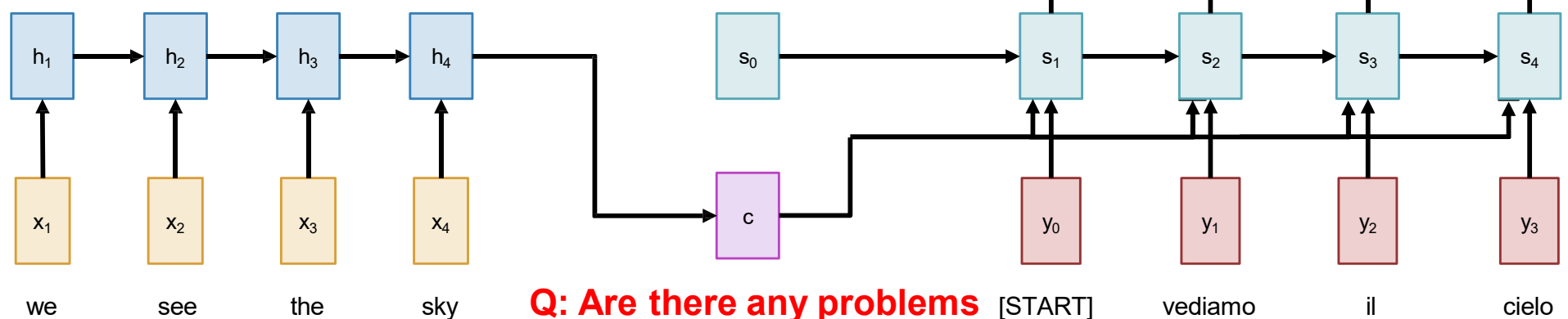
Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$

Encoder: $h_t = f_W(x_t, h_{t-1})$

From final hidden state predict:

Initial decoder state s_0

Context vector c (often $c=h_T$)



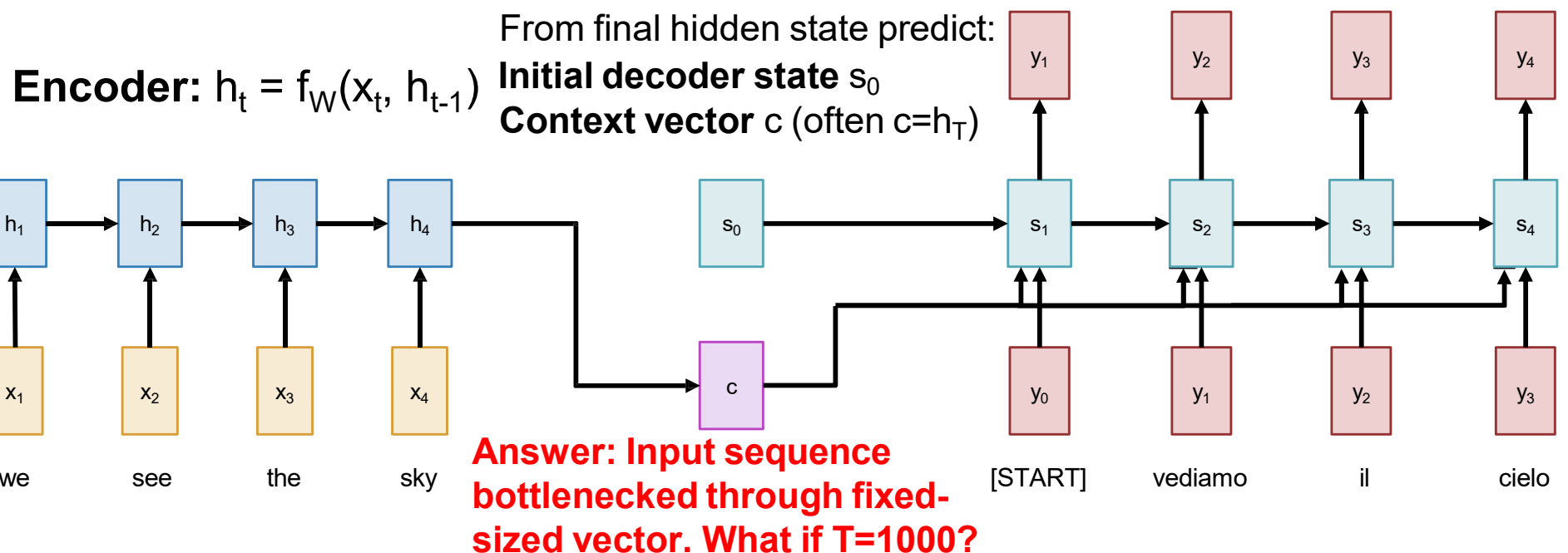
Q: Are there any problems with using C like this??

Sequence to Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$



Sequence to Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

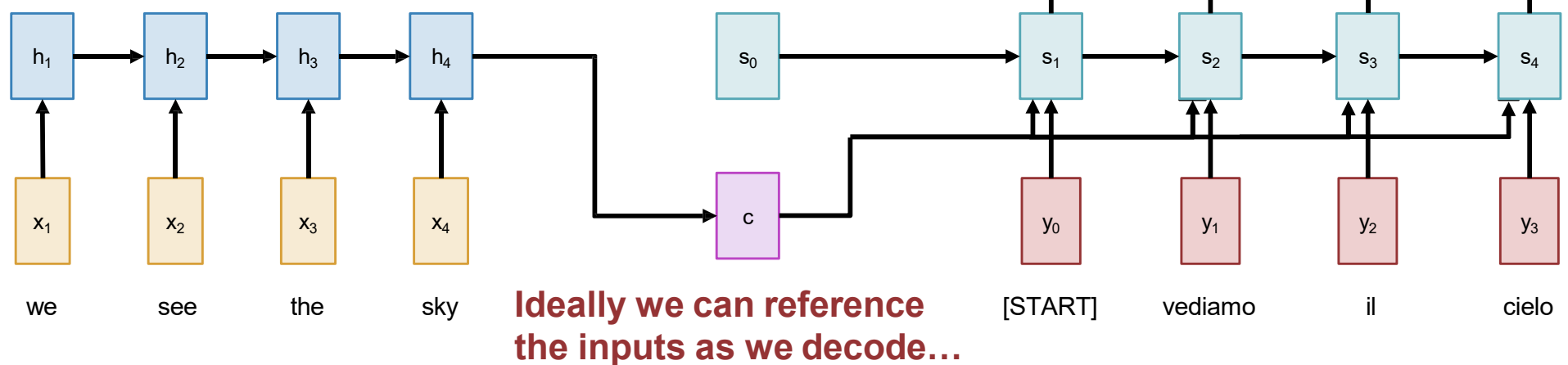
Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$

Encoder: $h_t = f_W(x_t, h_{t-1})$

From final hidden state predict:

Initial decoder state s_0

Context vector c (often $c=h_T$)

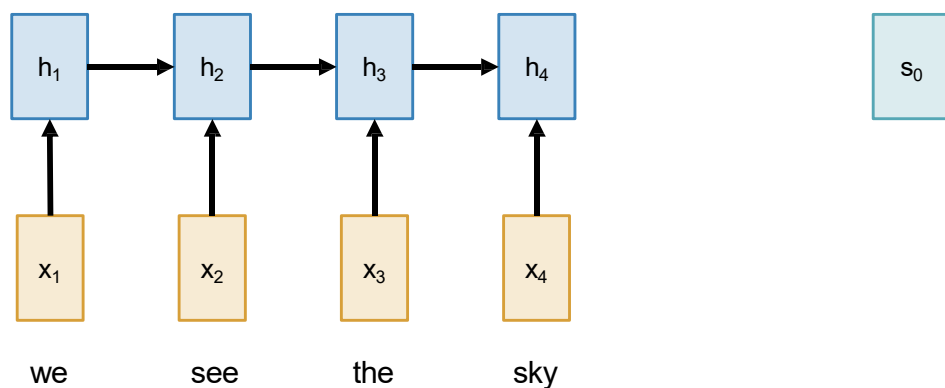


Sequence to Sequence with RNNs and Attention

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_T$

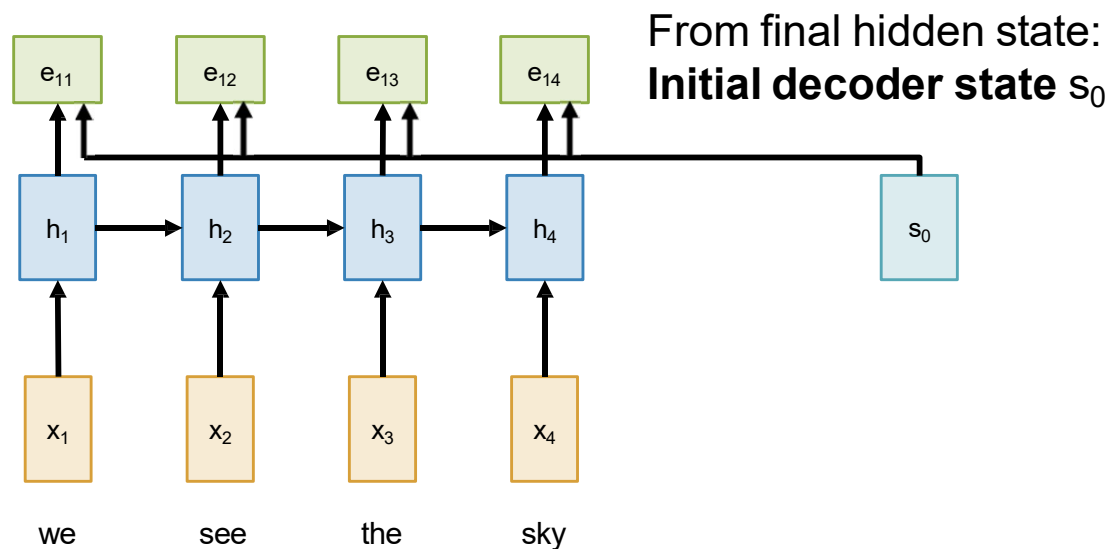
Encoder: $h_t = f_W(x_t, h_{t-1})$ From final hidden state:
Initial decoder state s_0



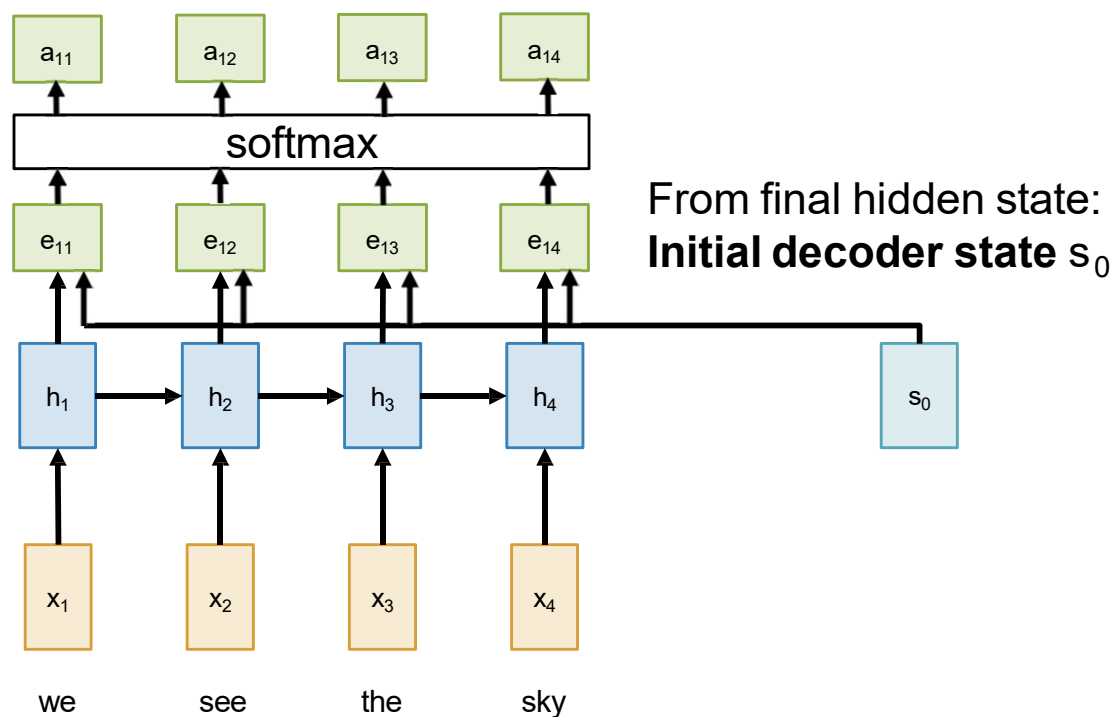
Sequence to Sequence with RNNs and **Attention**

Compute (scalar) **alignment scores**

$$e_{t,i} = f_{\text{att}}(s_{t-1}, h_i) \quad (f_{\text{att}} \text{ is a Linear Layer})$$



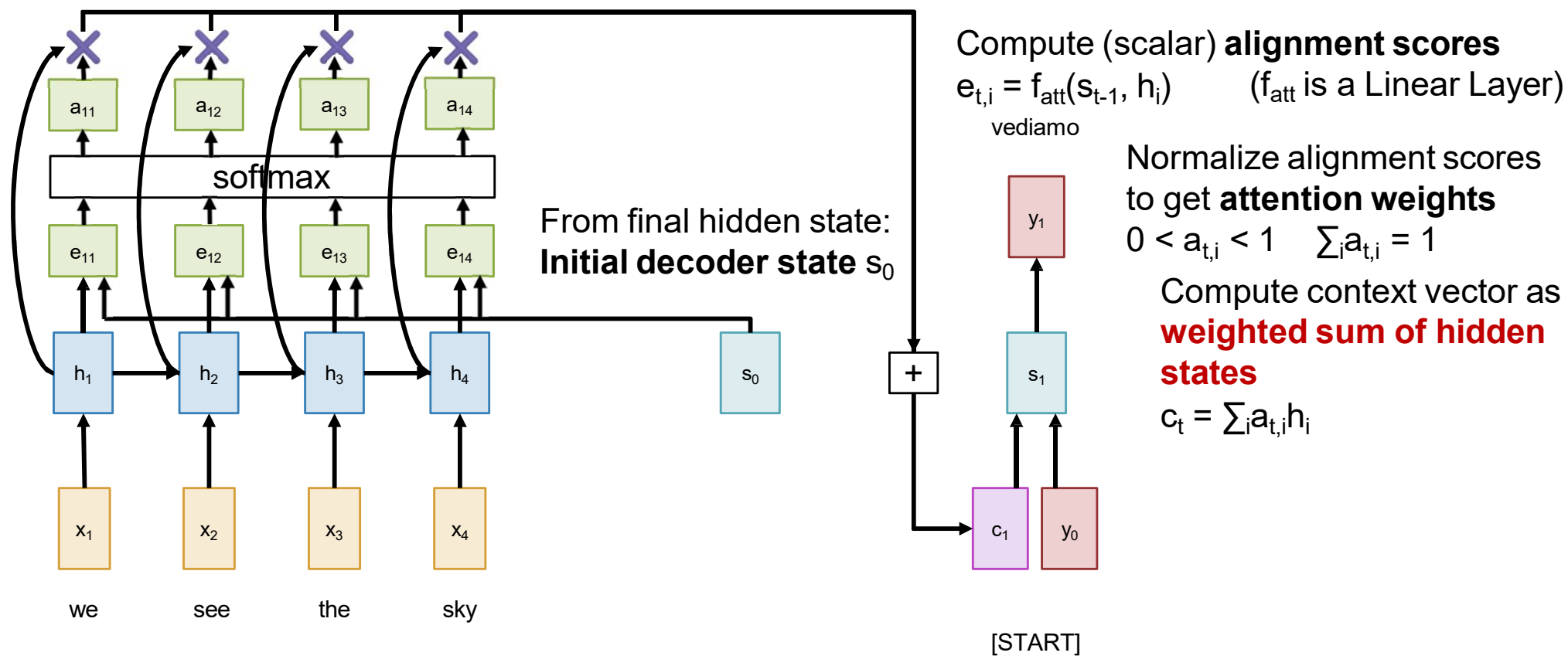
Sequence to Sequence with RNNs and **Attention**



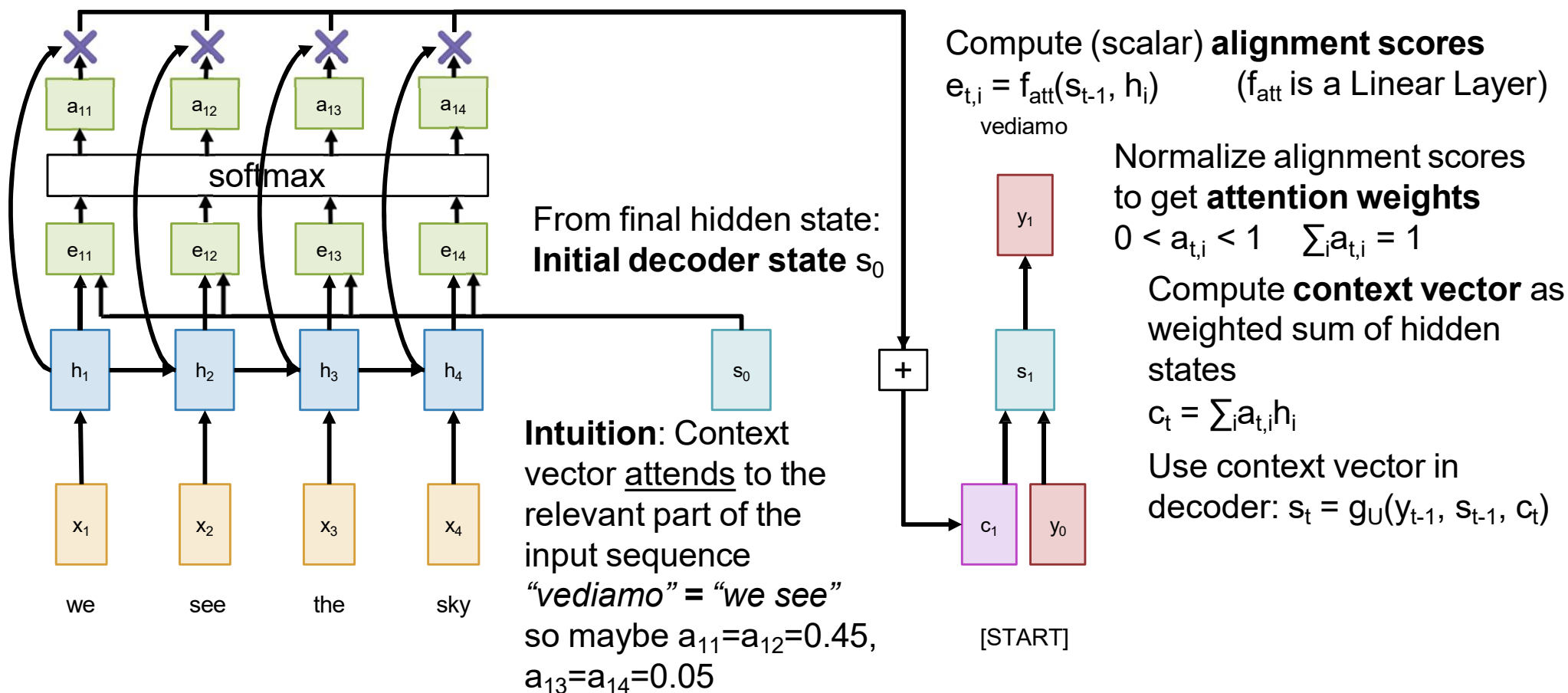
Compute (scalar) **alignment scores**
 $e_{t,i} = f_{\text{att}}(s_{t-1}, h_i)$ (f_{att} is a Linear Layer)

Normalize alignment scores
to get **attention weights**
 $0 < a_{t,i} < 1 \quad \sum_i a_{t,i} = 1$

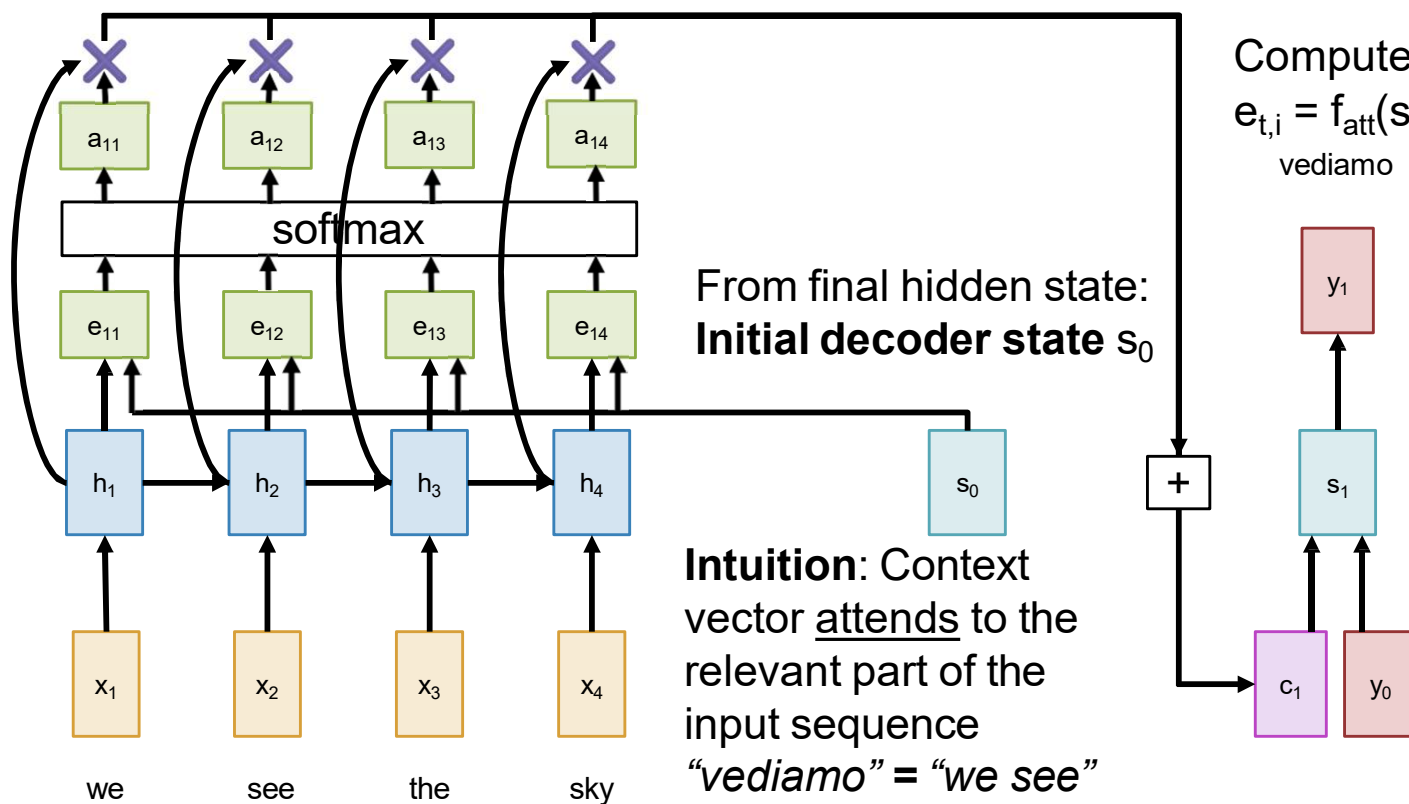
Sequence to Sequence with RNNs and Attention



Sequence to Sequence with RNNs and Attention



Sequence to Sequence with RNNs and **Attention**



From final hidden state:
Initial decoder state s_0

Intuition: Context vector attends to the relevant part of the input sequence
"vediamo" = "we see"
so maybe $a_{11}=a_{12}=0.45$,
 $a_{13}=a_{14}=0.05$

Compute (scalar) **alignment scores**
 $e_{t,i} = f_{\text{att}}(s_{t-1}, h_i)$ (f_{att} is a Linear Layer)
vediamo

Normalize alignment scores to get **attention weights**
 $0 < a_{t,i} < 1 \quad \sum_i a_{t,i} = 1$

Compute **context vector** as weighted sum of hidden states

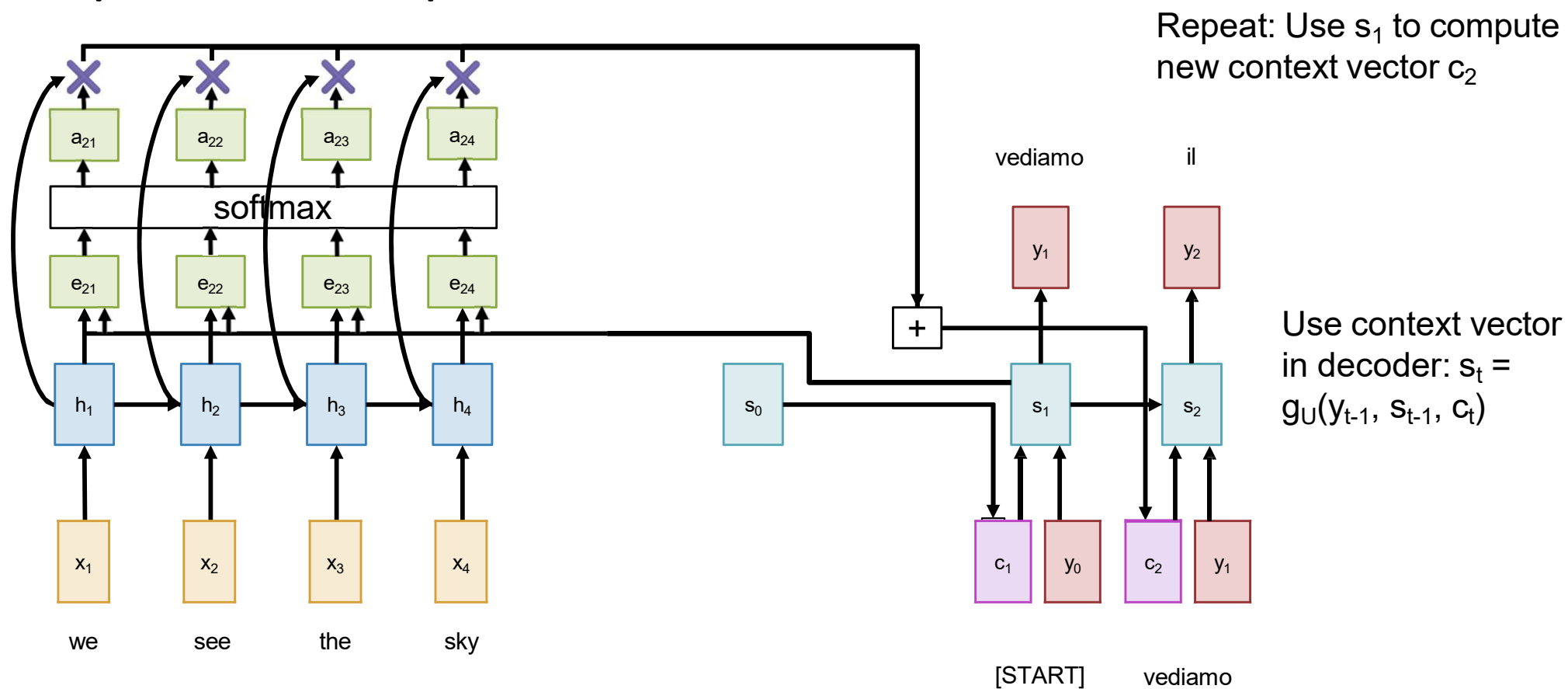
$$c_t = \sum_i a_{t,i} h_i$$

Use context vector in decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c_t)$

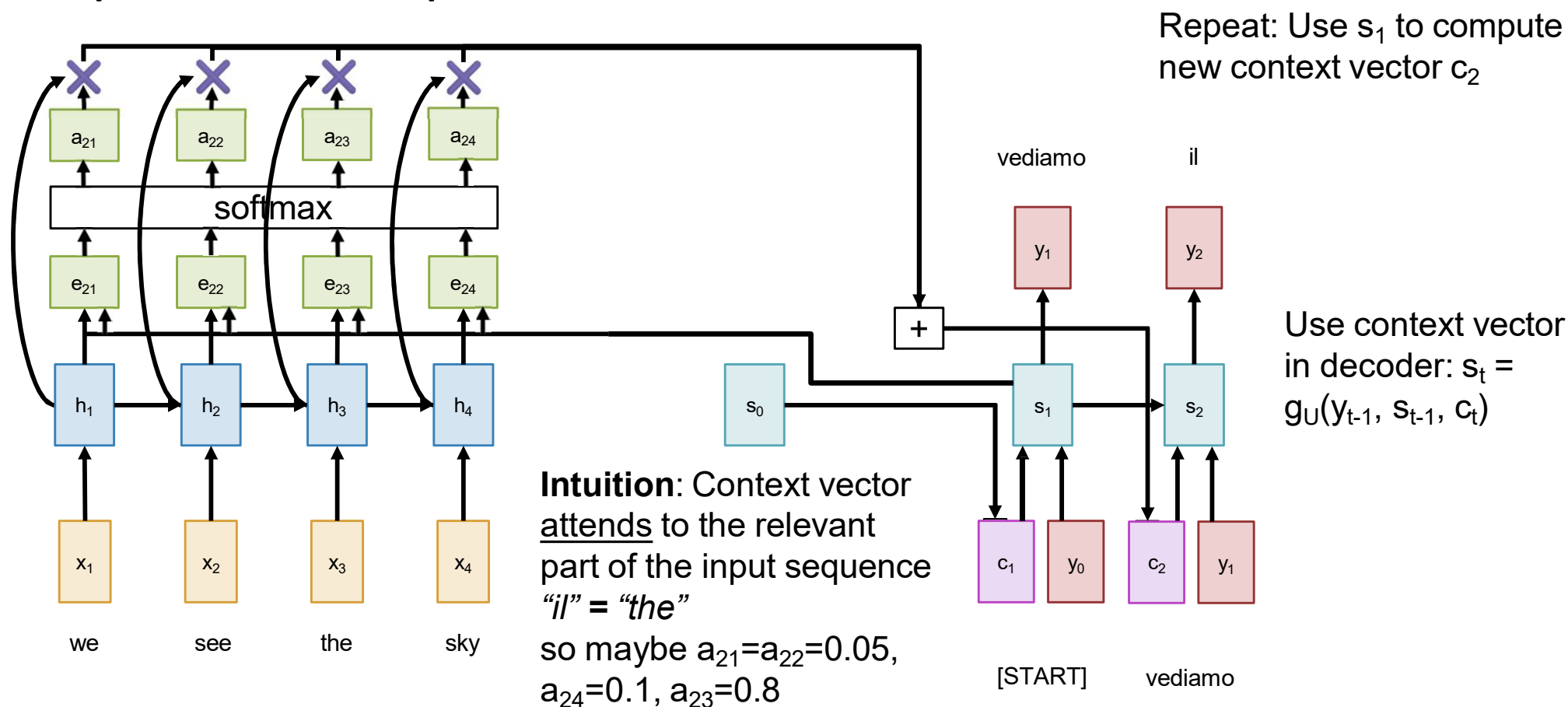
This is all differentiable! No supervision on attention weights – backprop through everything

[START]

Sequence to Sequence with RNNs and Attention



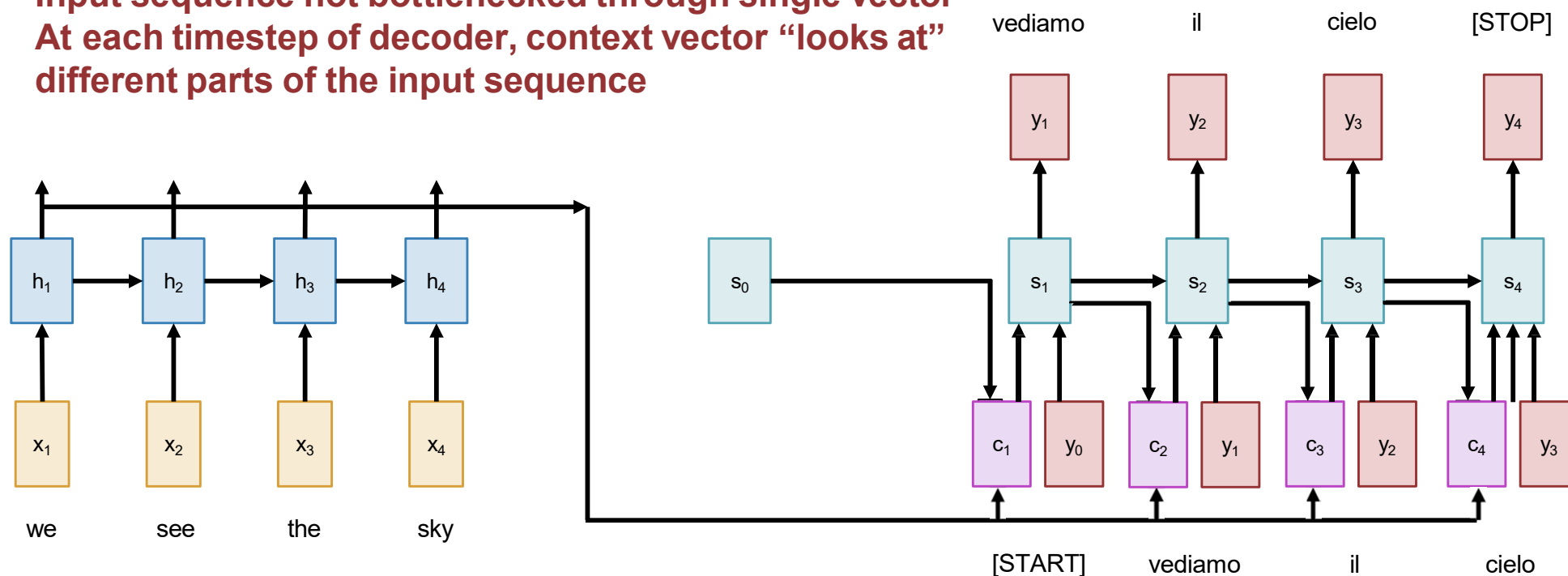
Sequence to Sequence with RNNs and Attention



Sequence to Sequence with RNNs and Attention

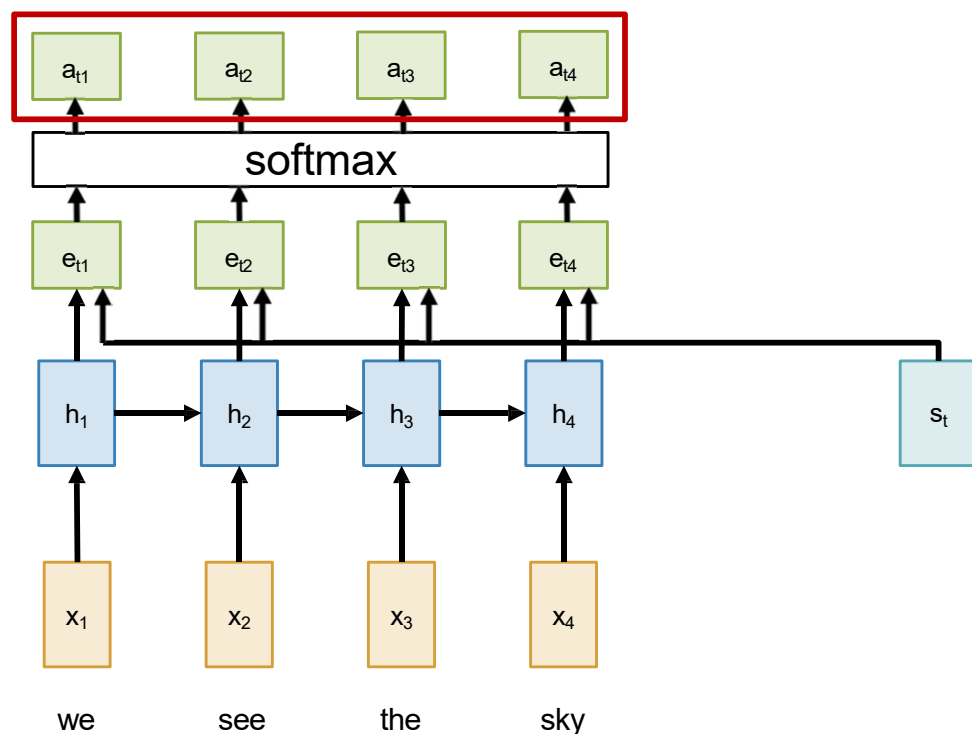
Use a different context vector in each timestep of decoder

- Input sequence not bottlenecked through single vector
- At each timestep of decoder, context vector “looks at” different parts of the input sequence

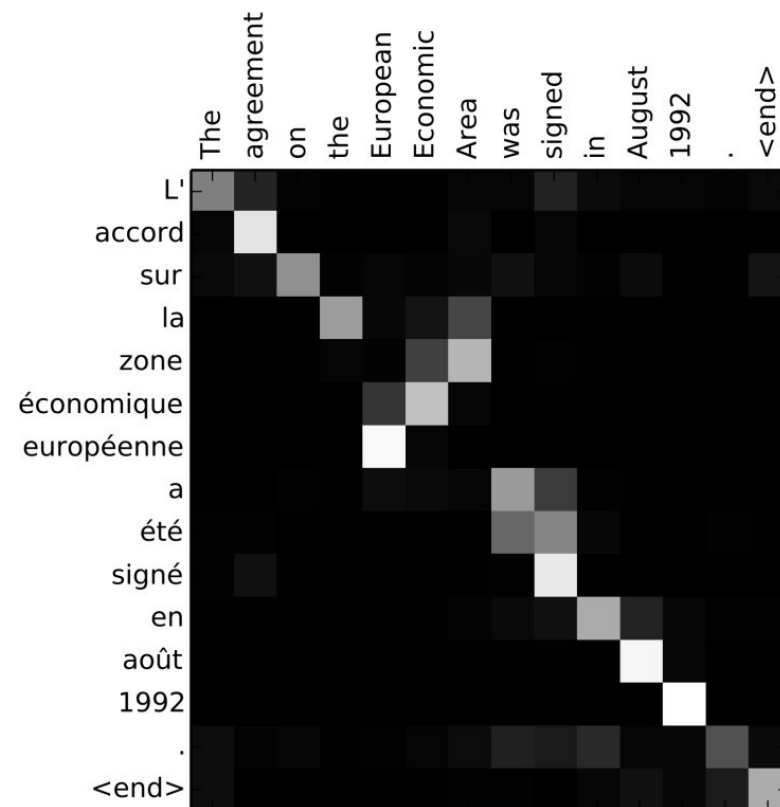


Sequence to Sequence with RNNs and Attention

Example: English to French translation



Visualize attention weights $a_{t,i}$



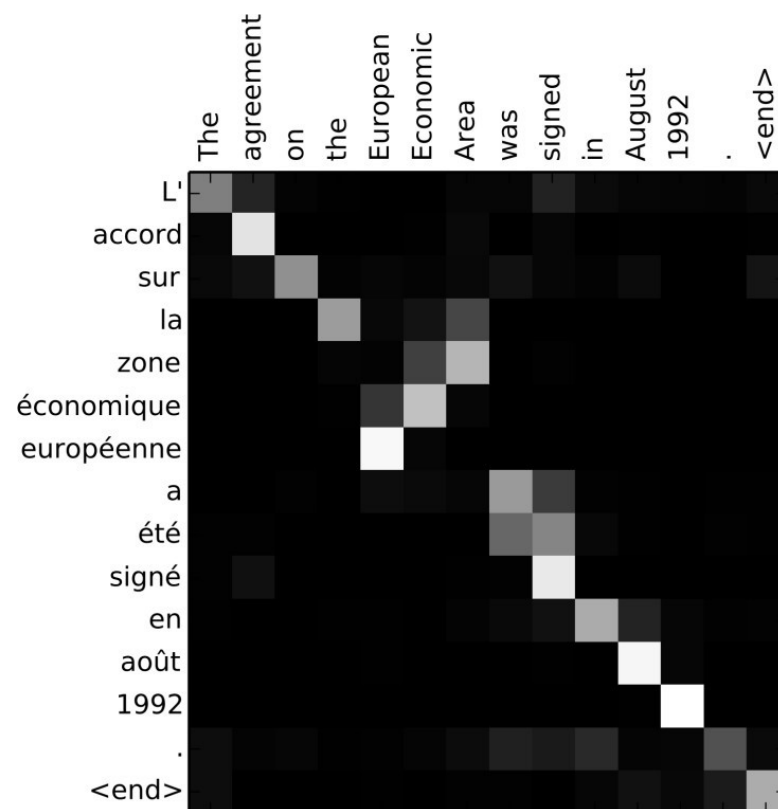
Sequence to Sequence with RNNs and Attention

Example: English to French translation

Input: “The agreement on the European Economic Area was signed in August 1992.”

Output: “L’accord sur la zone économique européenne a été signé en août 1992.”

Visualize attention weights $a_{t,i}$



Sequence to Sequence with RNNs and Attention

Example: English to French translation

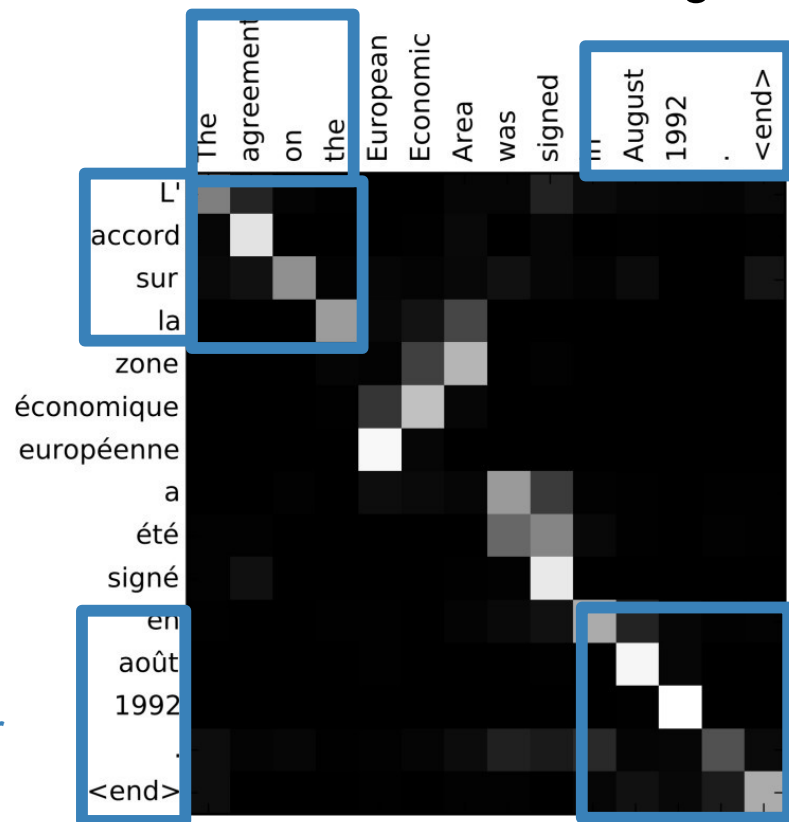
Input: “**The agreement on the** European Economic Area was signed **in August 1992.**”

Output: “**L’accord sur la** zone économique européenne a été signé **en août 1992.**”

Diagonal attention means words correspond in order

Diagonal attention means words correspond in order

Visualize attention weights $a_{t,i}$



Sequence to Sequence with RNNs and Attention

Example: English to French translation

Input: “The agreement on the European Economic Area was signed in August 1992.”

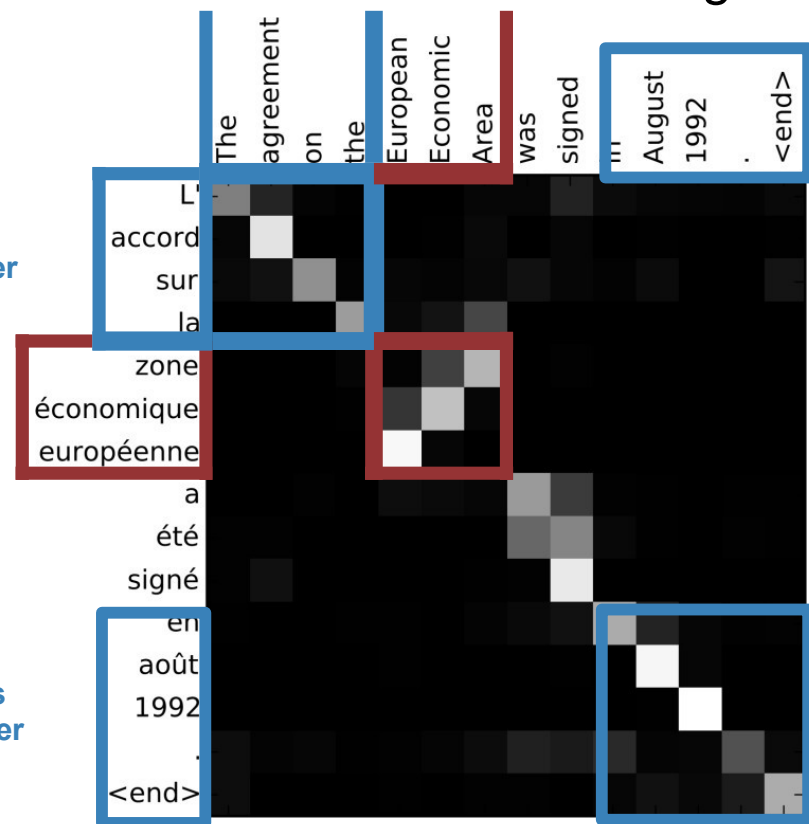
Output: “L’accord sur la zone économique européenne a été signé en août 1992.”

Diagonal attention means words correspond in order

Attention figures out different word orders

Diagonal attention means words correspond in order

Visualize attention weights $a_{t,i}$



Sequence to Sequence with RNNs and Attention

Context vectors don't use the fact that h_i form an ordered sequence – it just treats them as an unordered set $\{h_i\}$

General architecture + strategy given any set of input hidden vectors $\{h_i\}$! (calculate attention weights + sum)

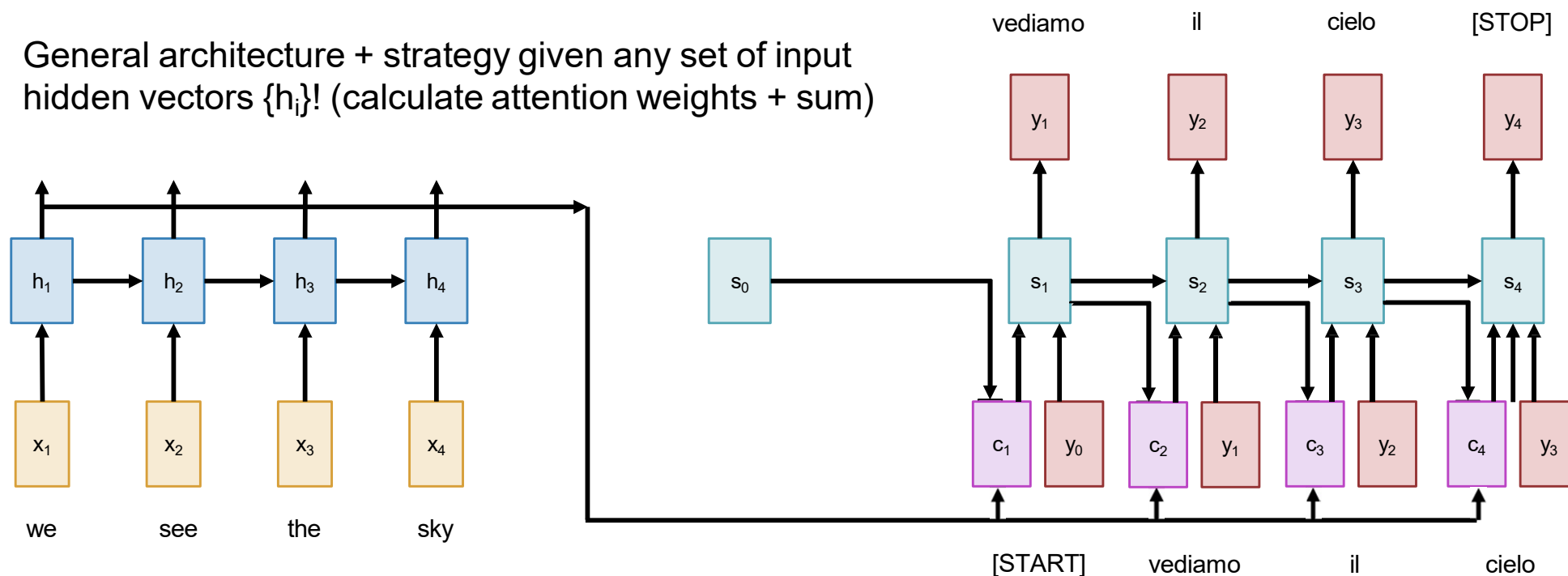


Image Captioning using spatial features

Input: Image I

Output: Sequence $\mathbf{y} = y_1, y_2, \dots, y_T$

An example network for image captioning
without attention

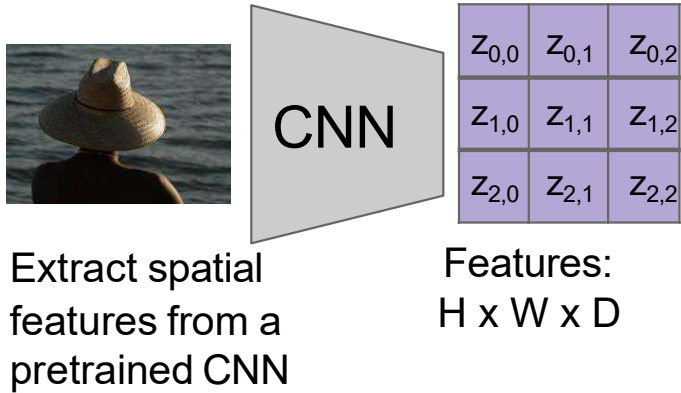


Image Captioning using spatial features

Input: Image I

Output: Sequence $\mathbf{y} = y_1, y_2, \dots, y_T$

Encoder: $h_0 = f_w(\mathbf{z})$

where $\mathbf{z}$ is spatial CNN features

$f_w(\cdot)$ is an MLP

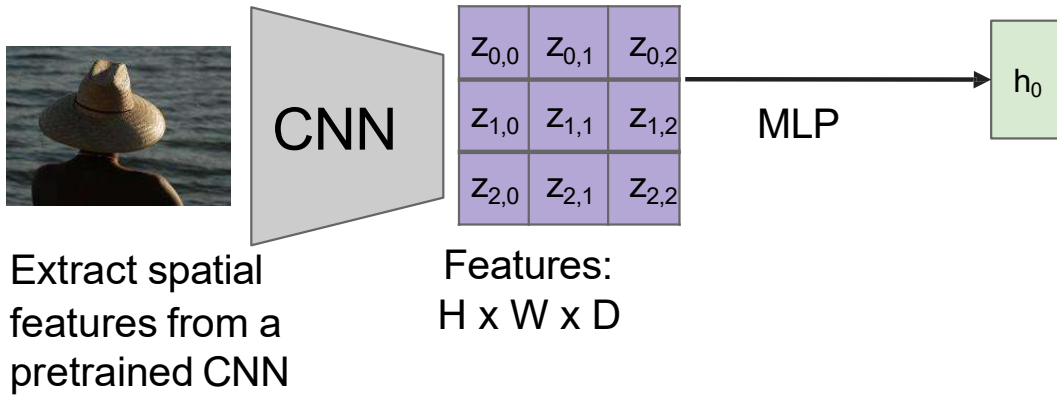


Image Captioning using spatial features

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where $\mathbf{z}$ is spatial CNN features
 $f_w(\cdot)$ is an MLP

Decoder: $h_t = g_v(y_{t-1}, h_{t-1}, c)$
where context vector c is often $c = h_0$
and output $y_t = T(h_t)$

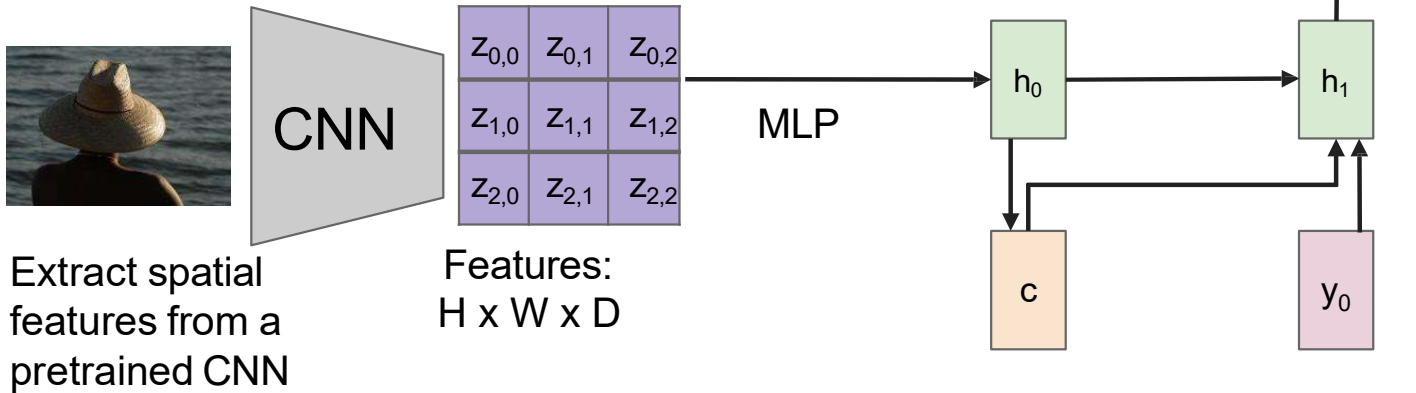
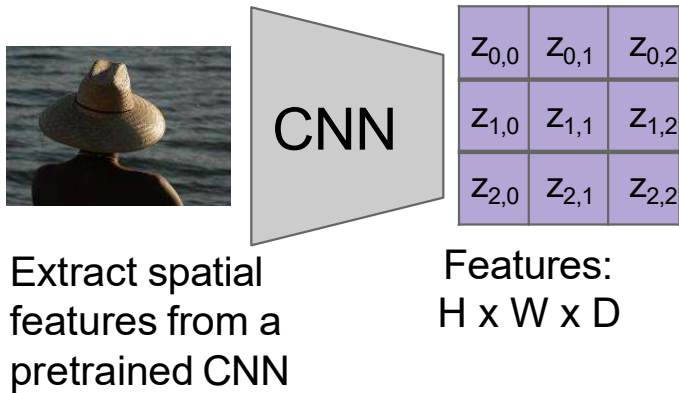


Image Captioning using spatial features

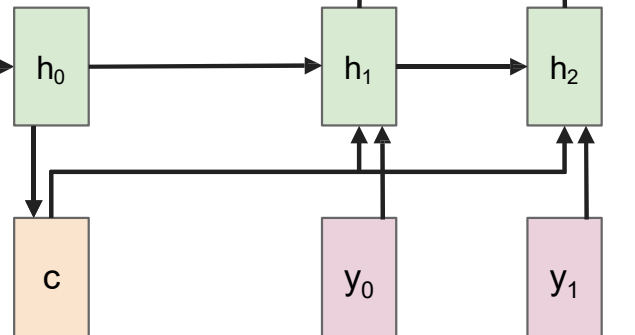
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Encoder: $h_0 = f_w(\mathbf{z})$
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 $f_w(\cdot)$ is an MLP



MLP



Decoder: $h_t = g_v(y_{t-1}, h_{t-1}, c)$
where context vector c is often $c = h_0$
and output $y_t = T(h_t)$

person

wearing

[START]

person

Image Captioning using spatial features

Input: Image I

Output: Sequence $\mathbf{y} = y_1, y_2, \dots, y_T$

Encoder: $h_0 = f_w(\mathbf{z})$
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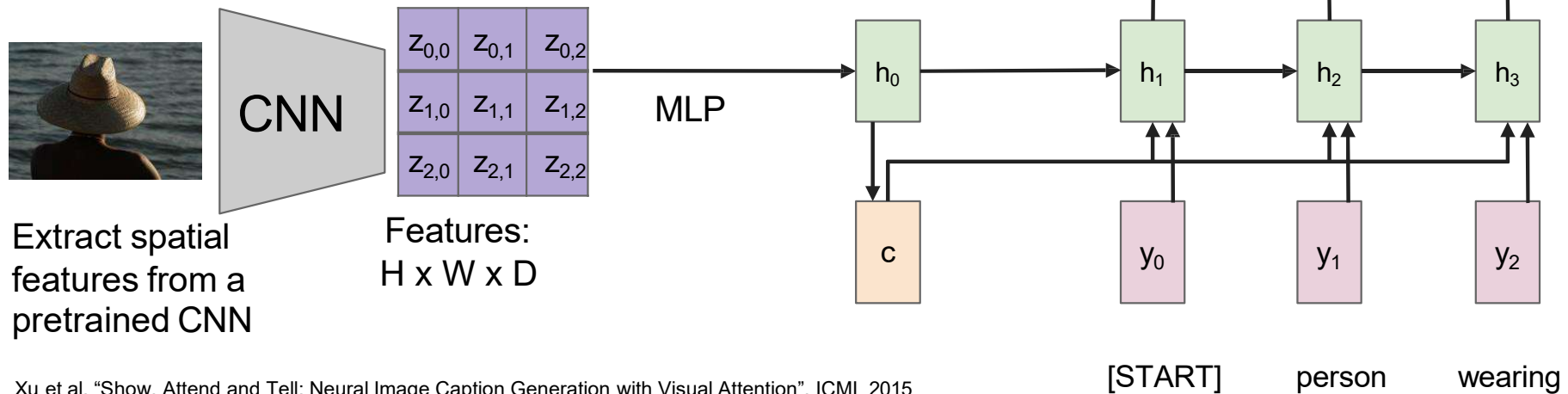
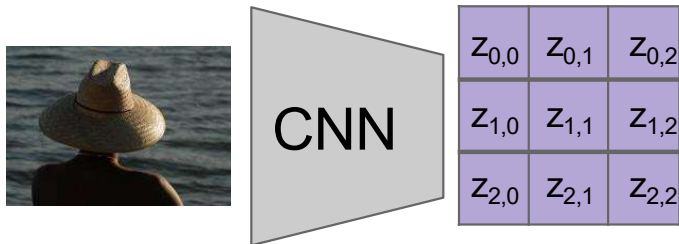


Image Captioning using spatial features

Input: Image I

Output: Sequence $\mathbf{y} = y_1, y_2, \dots, y_T$

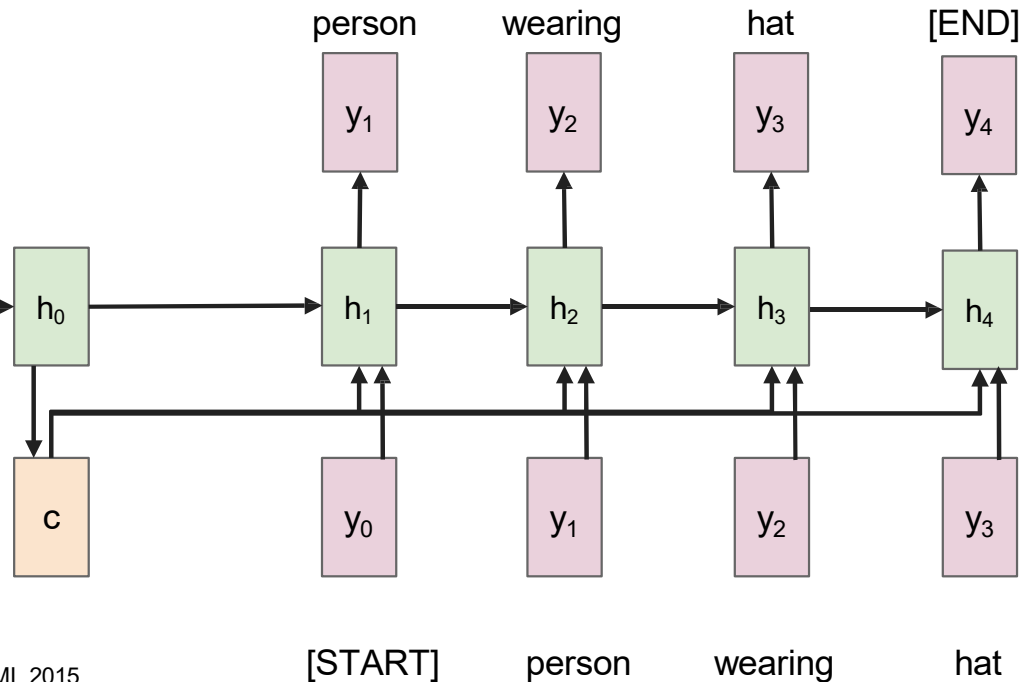
Encoder: $h_0 = f_w(\mathbf{z})$
where $\mathbf{z}$ is spatial CNN features
 $f_w(\cdot)$ is an MLP



Extract spatial features from a pretrained CNN

Features:
 $H \times W \times D$

MLP



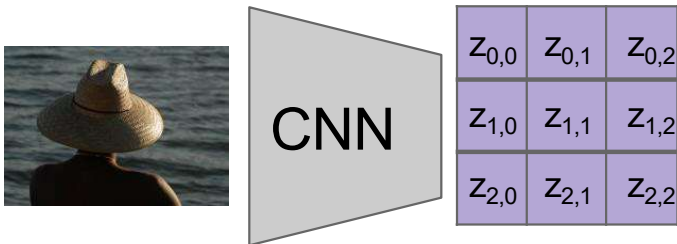
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Image Captioning using spatial features

Input: Image I

Output: Sequence $\mathbf{y} = y_1, y_2, \dots, y_T$

Encoder: $h_0 = f_w(\mathbf{z})$
where $\mathbf{z}$ is spatial CNN features
 $f_w(\cdot)$ is an MLP



Extract spatial features from a pretrained CNN

Q: What is the problem with this setup? Think back to last time...

Decoder: $h_t = g_v(y_{t-1}, h_{t-1}, c)$
where context vector c is often $c = h_0$
and output $y_t = T(h_t)$

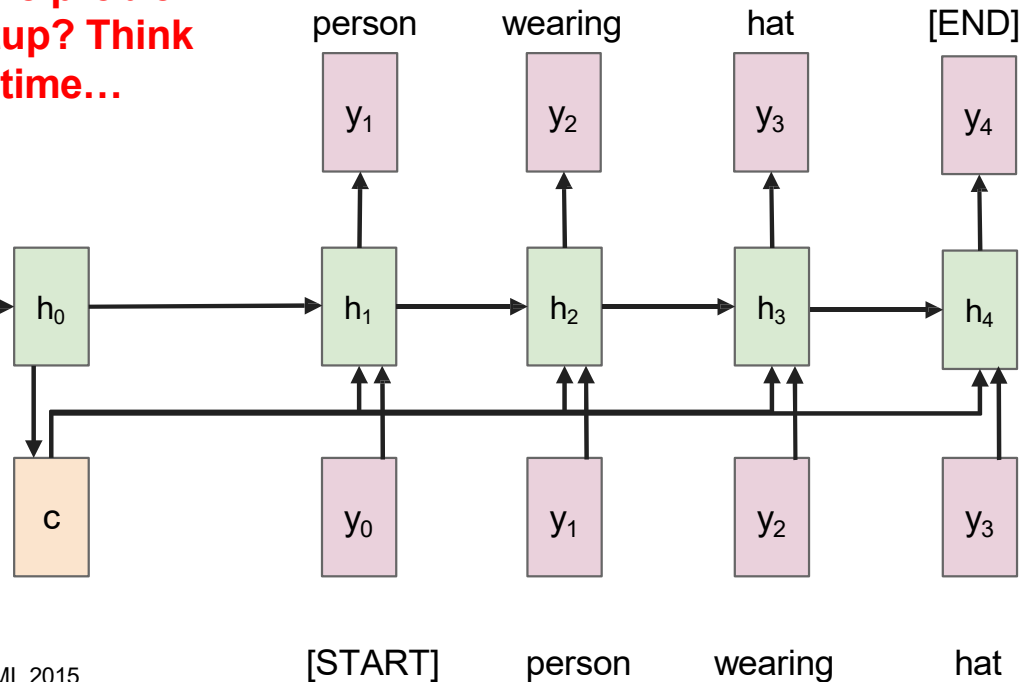


Image Captioning using spatial features

Answer: Input is "bottlenecked" through c

- Model needs to encode everything it wants to say within c

This is a problem if we want to generate really long descriptions? 100s of words long

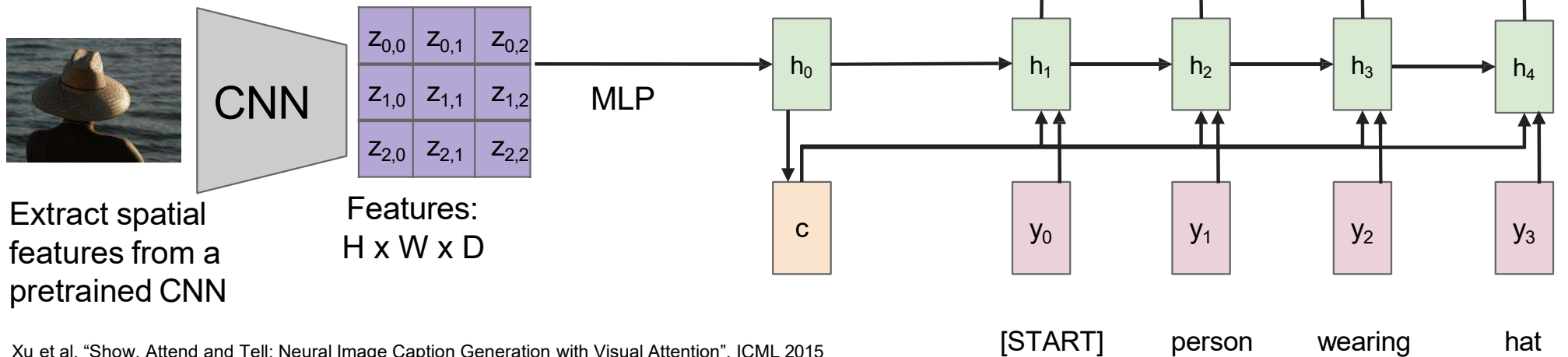
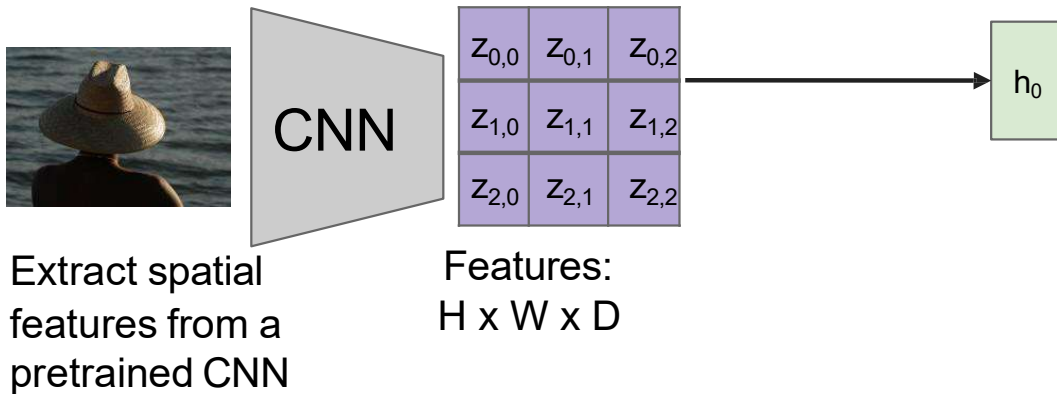


Image Captioning with RNNs and Attention

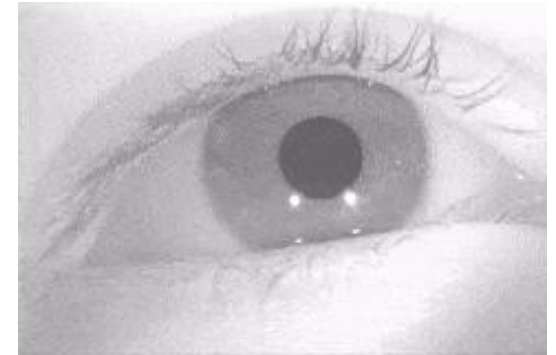
Attention idea: New context vector at every time step.

Each context vector will attend to different image regions



Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

[gif source](#)



Attention Saccades in humans

Image Captioning with RNNs and Attention

Compute alignments scores (scalars):

$$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$$

$f_{att}(\cdot)$ is an MLP

Alignment scores:
H x W

$e_{1,0,0}$	$e_{1,0,1}$	$e_{1,0,2}$
$e_{1,1,0}$	$e_{1,1,1}$	$e_{1,1,2}$
$e_{1,2,0}$	$e_{1,2,1}$	$e_{1,2,2}$



CNN

Extract spatial features from a pretrained CNN

$z_{0,0}$	$z_{0,1}$	$z_{0,2}$
$z_{1,0}$	$z_{1,1}$	$z_{1,2}$
$z_{2,0}$	$z_{2,1}$	$z_{2,2}$

Features:
H x W x D

h_0

Image Captioning with RNNs and Attention

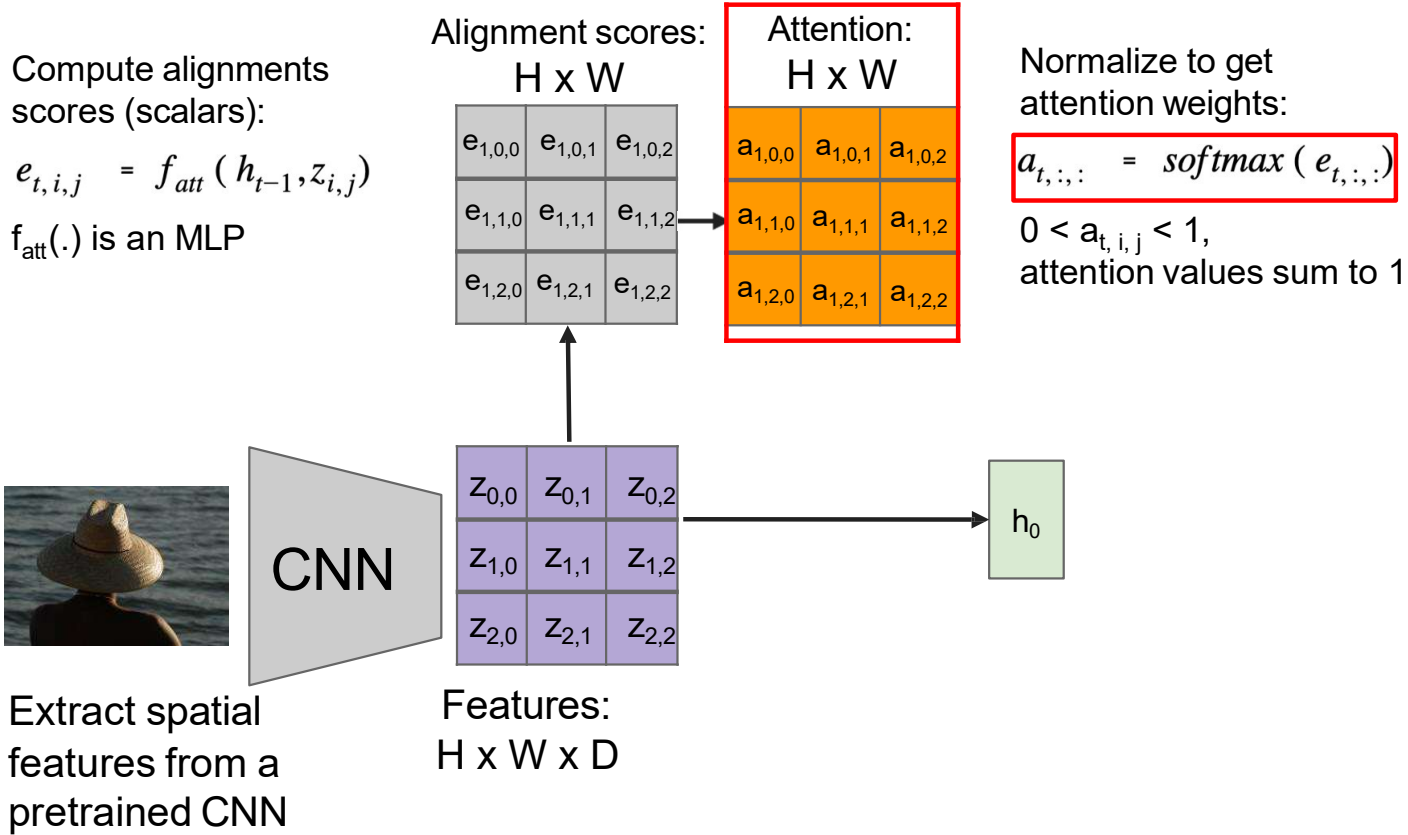


Image Captioning with RNNs and Attention

Compute alignments scores (scalars):

$$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$$

$f_{att}(\cdot)$ is an MLP

Alignment scores:
H x W

$e_{1,0,0}$	$e_{1,0,1}$	$e_{1,0,2}$
$e_{1,1,0}$	$e_{1,1,1}$	$e_{1,1,2}$
$e_{1,2,0}$	$e_{1,2,1}$	$e_{1,2,2}$

Attention:
H x W

$a_{1,0,0}$	$a_{1,0,1}$	$a_{1,0,2}$
$a_{1,1,0}$	$a_{1,1,1}$	$a_{1,1,2}$
$a_{1,2,0}$	$a_{1,2,1}$	$a_{1,2,2}$

Normalize to get attention weights:

$$a_{t,:,:) = \text{softmax}(e_{t,:,:})$$

$0 < a_{t,i,j} < 1$,
attention values sum to 1

Compute context vector:

$$c_t = \sum_{i,j} a_{t,i,j} z_{t,i,j}$$



CNN

Extract spatial features from a pretrained CNN

$z_{0,0}$	$z_{0,1}$	$z_{0,2}$
$z_{1,0}$	$z_{1,1}$	$z_{1,2}$
$z_{2,0}$	$z_{2,1}$	$z_{2,2}$

Features:
H x W x D

h_0



c_1

Q: How many context vectors are computed?

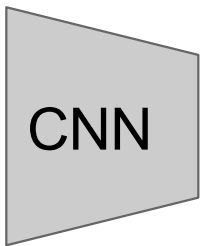
Image Captioning with RNNs and Attention

Each timestep of decoder uses a different context vector that looks at different parts of the input image

$$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$$

$$a_{t,:,:) = \text{softmax}(e_{t,:,:})$$

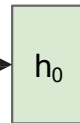
$$c_t = \sum_{i,j} a_{t,i,j} z_{t,i,j}$$



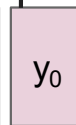
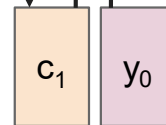
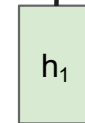
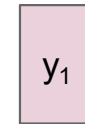
Extract spatial features from a pretrained CNN

$z_{0,0}$	$z_{0,1}$	$z_{0,2}$
$z_{1,0}$	$z_{1,1}$	$z_{1,2}$
$z_{2,0}$	$z_{2,1}$	$z_{2,2}$

Features:
H x W x D



person



[START]

$$\text{Decoder: } y_t = g_v(y_{t-1}, h_{t-1}, c_t)$$

New context vector at every time step

Image Captioning with RNNs and Attention

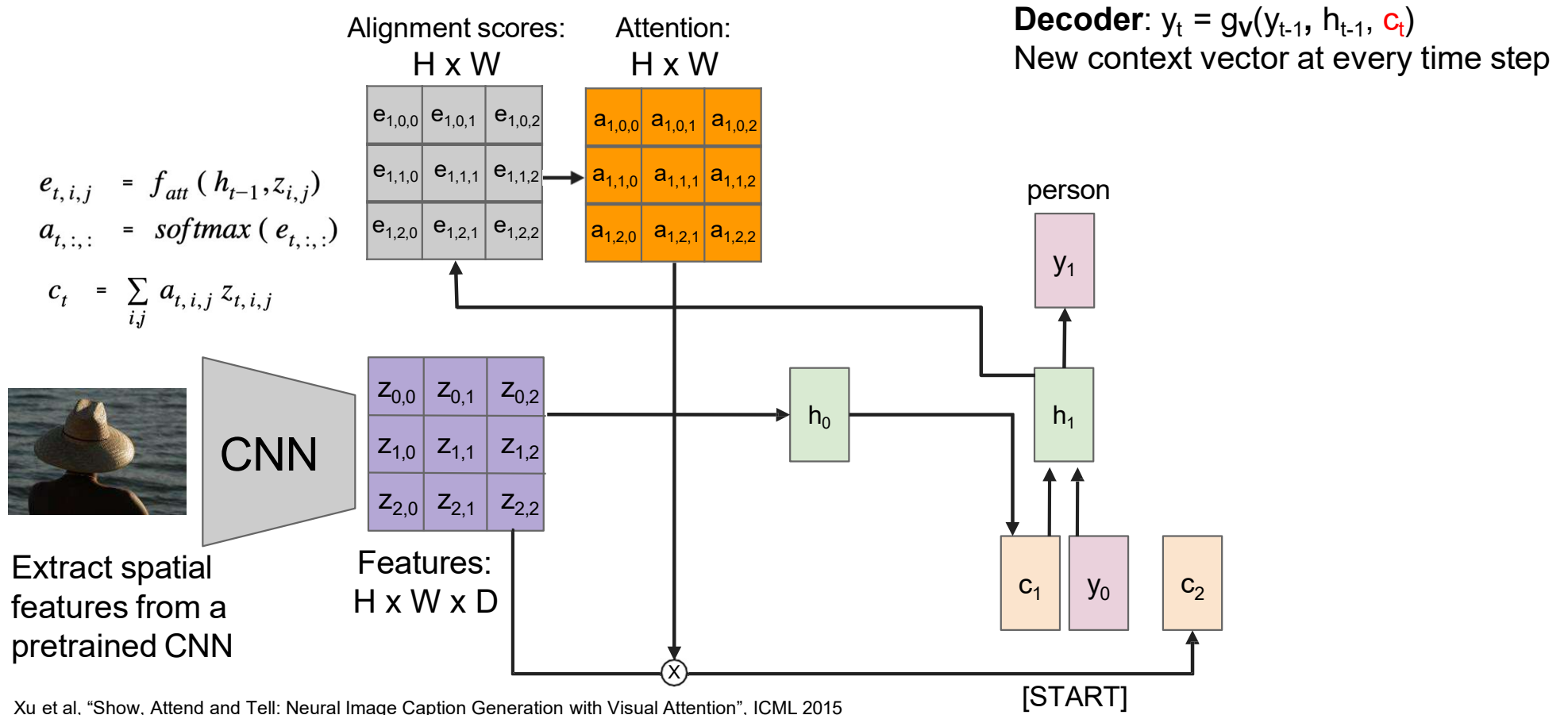


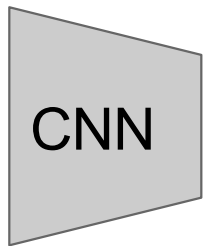
Image Captioning with RNNs and Attention

Each timestep of decoder uses a different context vector that looks at different parts of the input image

$$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$$

$$a_{t,:,:) = softmax(e_{t,:,:})$$

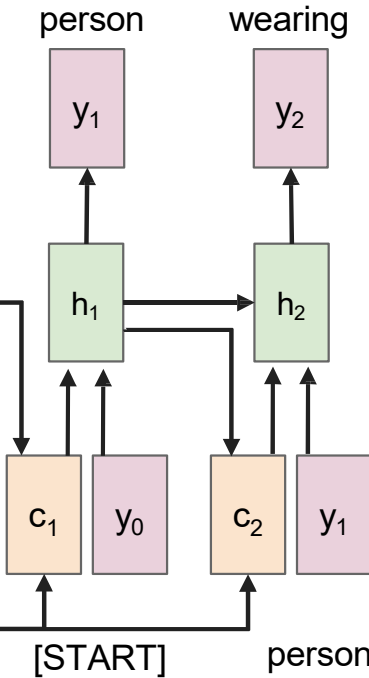
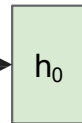
$$c_t = \sum_{i,j} a_{t,i,j} z_{t,i,j}$$



Extract spatial features from a pretrained CNN

$z_{0,0}$	$z_{0,1}$	$z_{0,2}$
$z_{1,0}$	$z_{1,1}$	$z_{1,2}$
$z_{2,0}$	$z_{2,1}$	$z_{2,2}$

Features:
H x W x D



Decoder: $y_t = g_v(y_{t-1}, h_{t-1}, c_t)$
New context vector at every time step

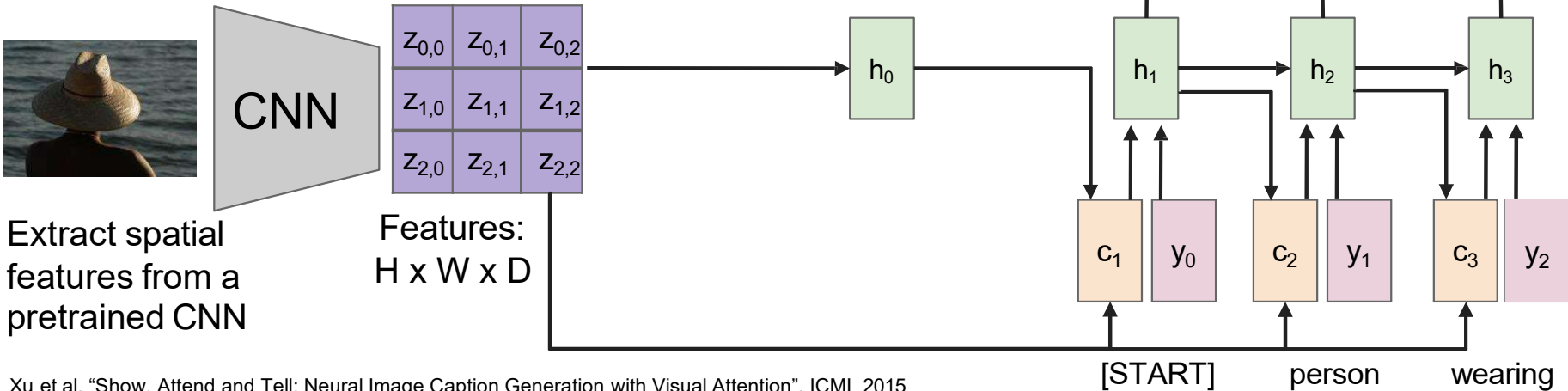
Image Captioning with RNNs and Attention

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Decoder: $y_t = g_v(y_{t-1}, h_{t-1}, c_t)$
New context vector at every time step

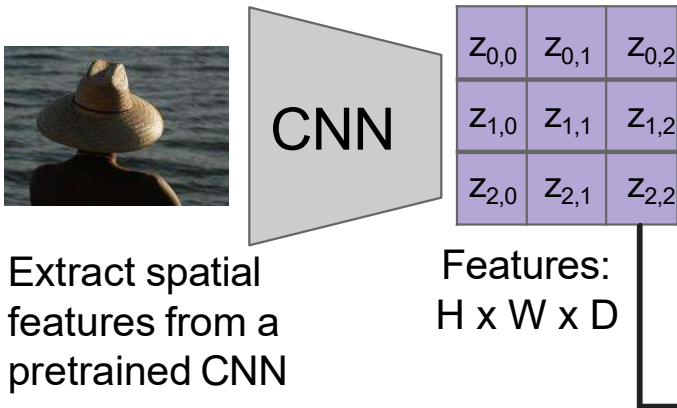
Image Captioning with RNNs and Attention

Each timestep of decoder uses a different context vector that looks at different parts of the input image

$$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$$

$$a_{t,:,:) = softmax(e_{t,:,:})$$

$$c_t = \sum_{i,j} a_{t,i,j} z_{t,i,j}$$



Decoder: $y_t = g_v(y_{t-1}, h_{t-1}, c_t)$
New context vector at every time step

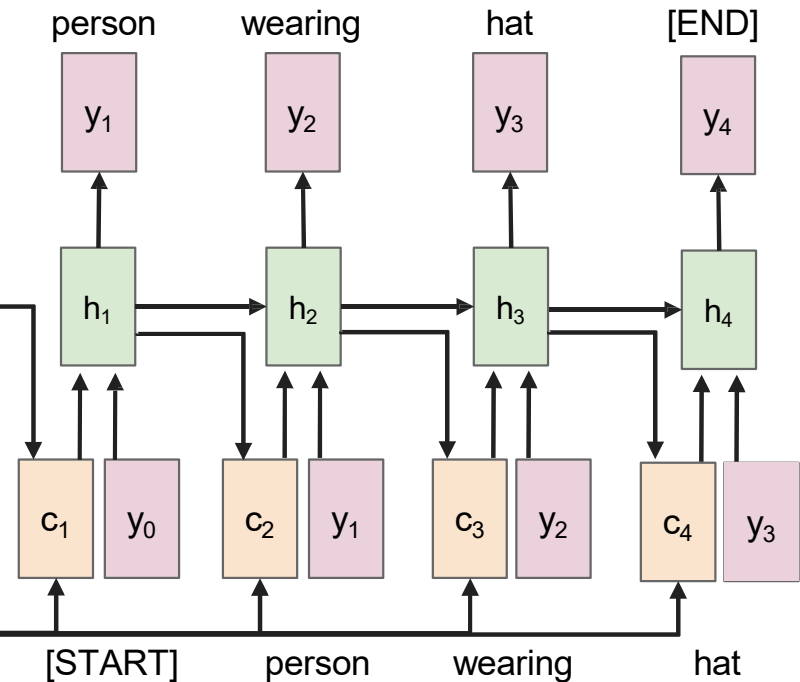


Image Captioning with RNNs and Attention

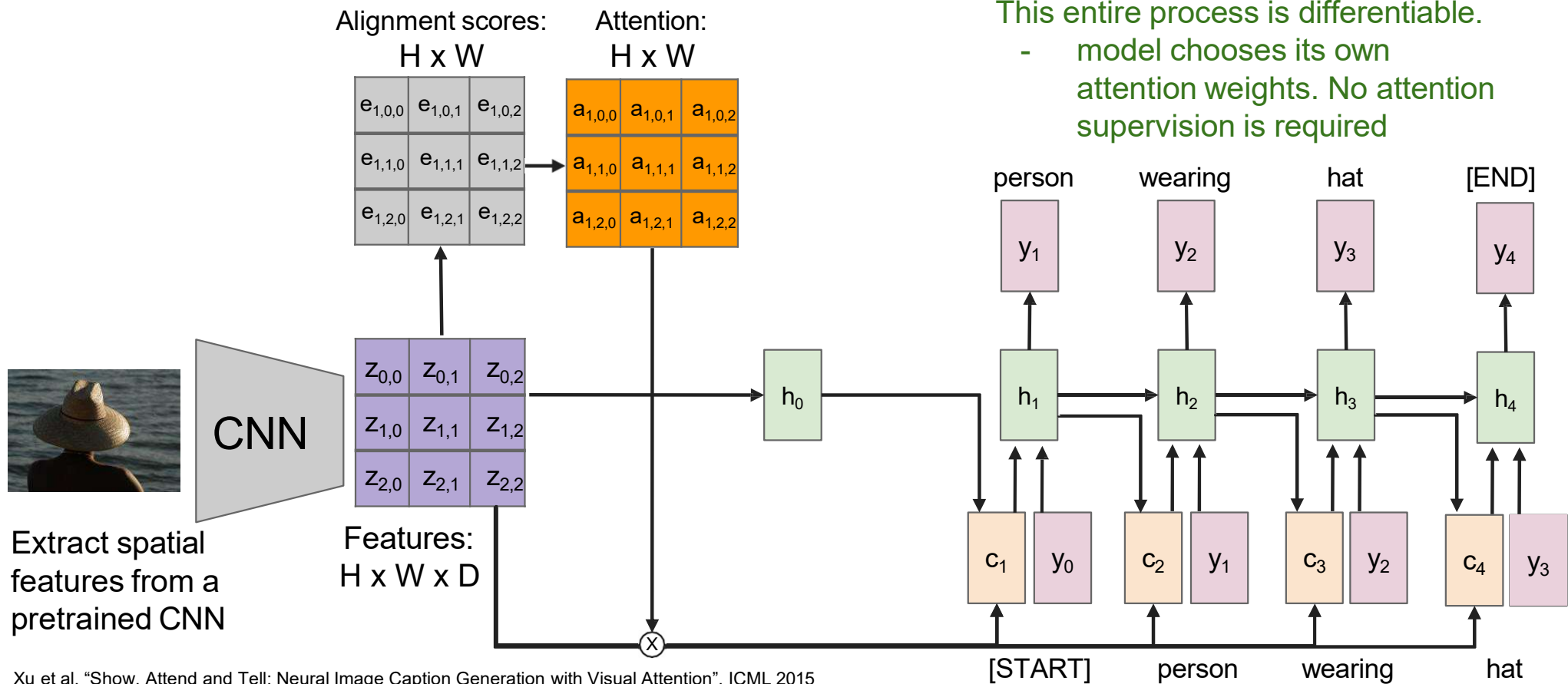
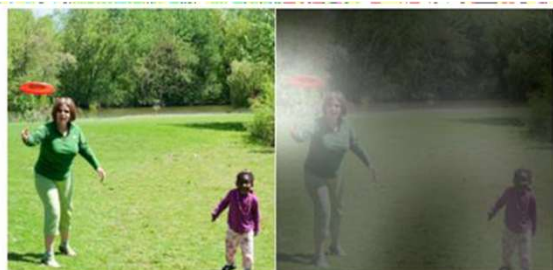


Image Captioning with Attention



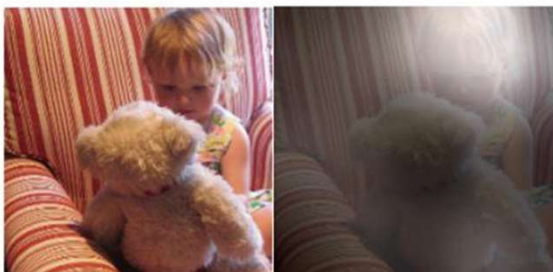
A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.

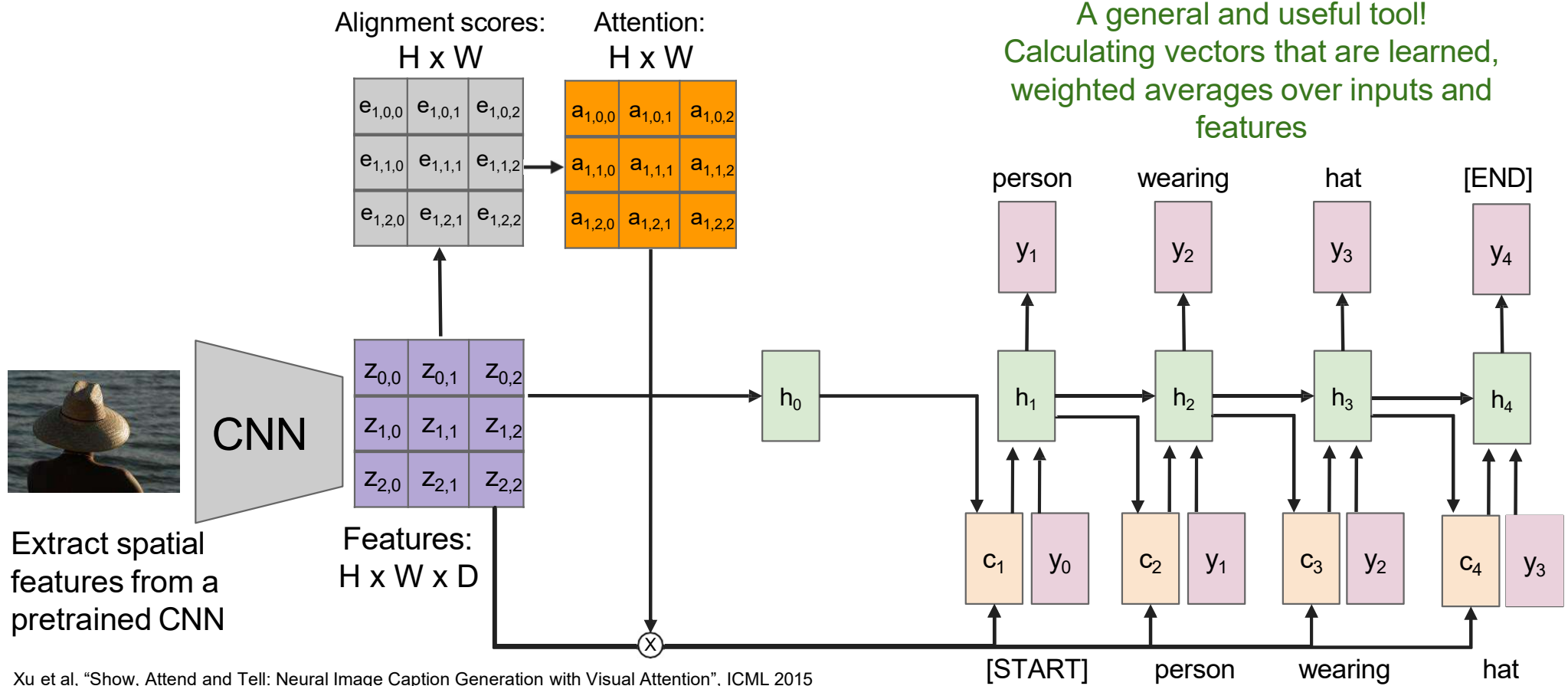


A giraffe standing in a forest with trees in the background.

Xu et al, "Show, Attend, and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Figure copyright Kelvin Xu, Jimmy Lei Ba, Jamie Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard S. Zemel, and Yoshua Bengio, 2015. Reproduced with permission.

Image Captioning with RNNs and Attention



Attention we just saw in image captioning

Features

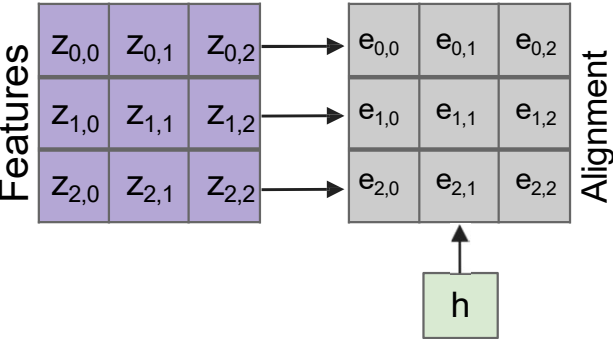
$z_{0,0}$	$z_{0,1}$	$z_{0,2}$
$z_{1,0}$	$z_{1,1}$	$z_{1,2}$
$z_{2,0}$	$z_{2,1}$	$z_{2,2}$

h

Inputs:
Features: $\mathbf{z}$ (shape: H x W x D)
Query: $\mathbf{h}$ (shape: D) $\leftarrow$ “query” refers to a vector used to calculate a corresponding context vector.

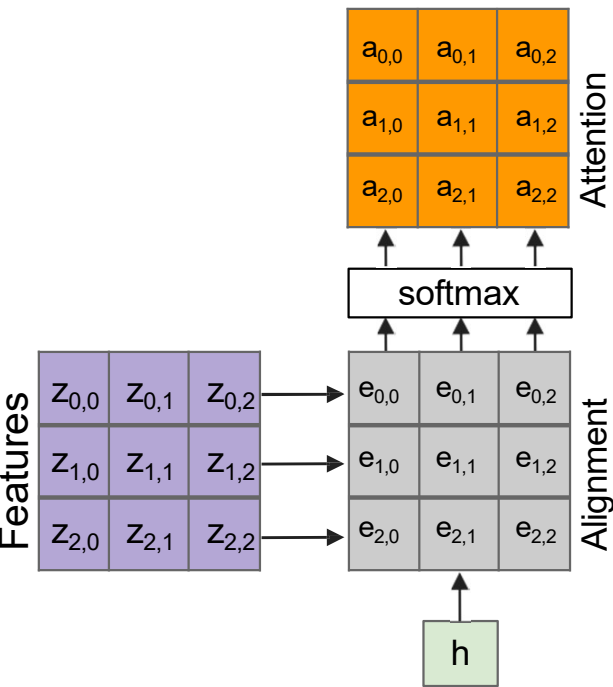
Attention we just saw in image captioning

Operations:
Alignment: $e_{i,j} = f_{\text{att}}(h, z_{i,j})$



Inputs:
Features: $\mathbf{z}$ (shape: $H \times W \times D$)
Query: $\mathbf{h}$ (shape: D)

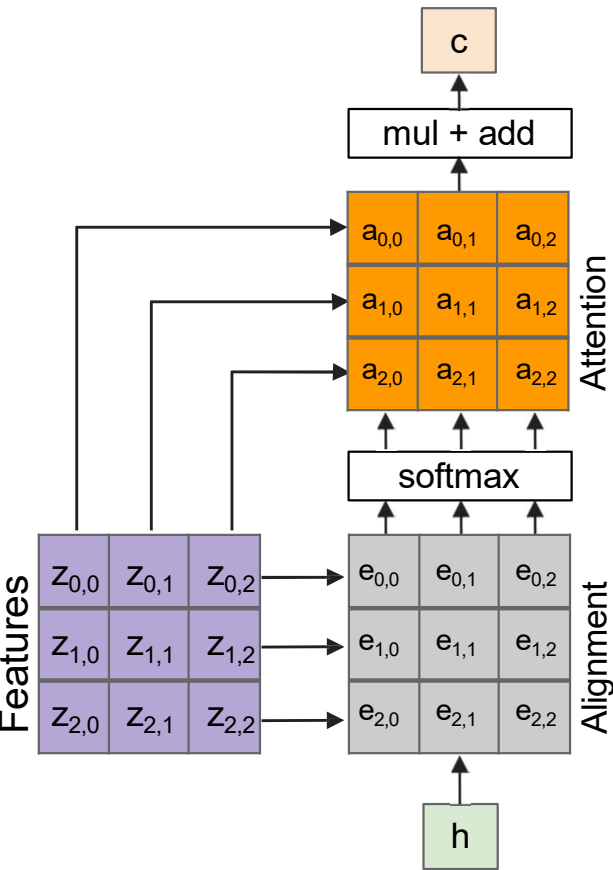
Attention we just saw in image captioning



Operations:
Alignment: $e_{i,j} = f_{\text{att}}(h, z_{i,j})$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$

Inputs:
Features: $\mathbf{z}$ (shape: $H \times W \times D$)
Query: $\mathbf{h}$ (shape: D)

Attention we just saw in image captioning

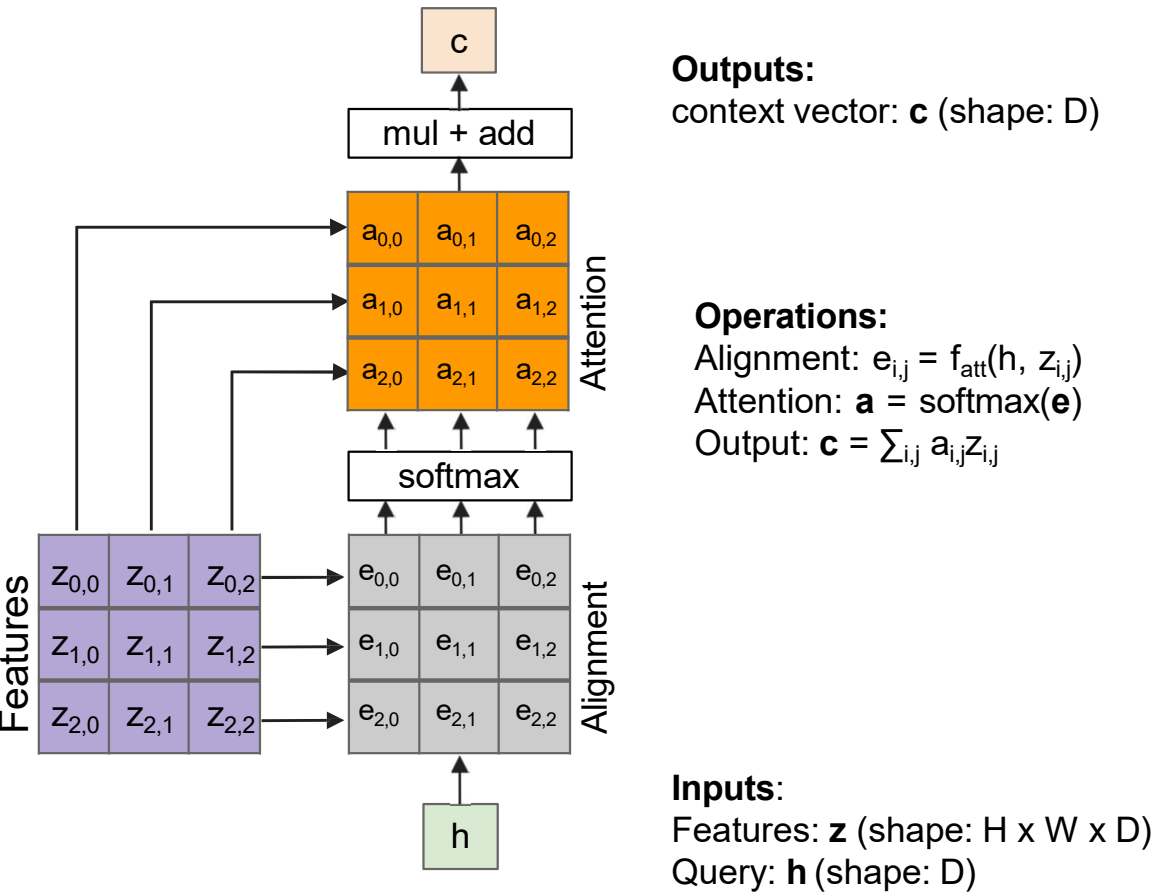


Outputs:
context vector: **c** (shape: D)

Operations:
Alignment: $e_{i,j} = f_{\text{att}}(h, z_{i,j})$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $\mathbf{c} = \sum_{i,j} a_{i,j} \mathbf{z}_{i,j}$

Inputs:
Features: **z** (shape: H x W x D)
Query: **h** (shape: D)

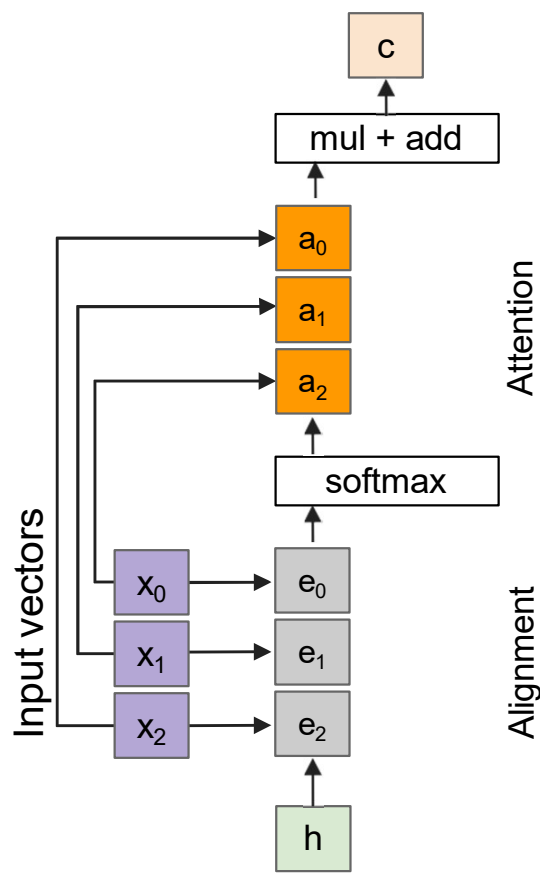
Attention we just saw in image captioning



How is this different from the attention mechanism in transformers?

We'll go into that next, any questions?

General attention layer – used in LLMs + beyond



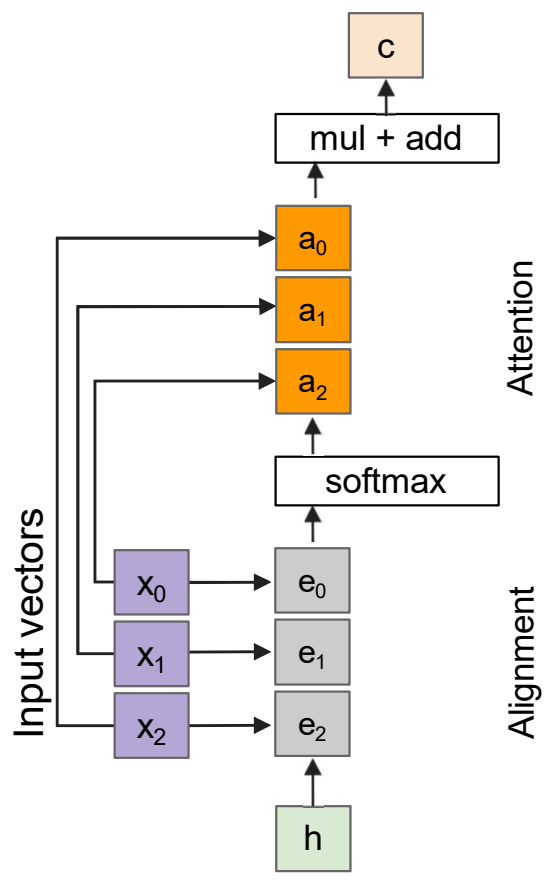
Outputs:
context vector: **c** (shape: D)

Operations:
Alignment: $e_i = f_{\text{att}}(h, x_i)$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $\mathbf{c} = \sum_i a_i x_i$

Inputs:
Input vectors: **x** (shape: N x D)
Query: **h** (shape: D)

- Attention operation is **permutation invariant**.
- Doesn't care about ordering of the features
 - Stretch into **N = H x W** vectors

General attention layer



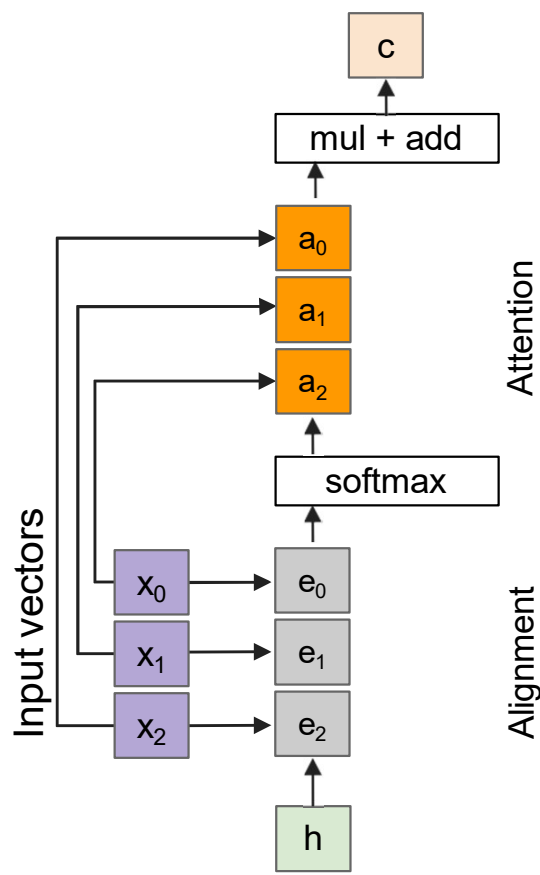
Outputs:
context vector: $\mathbf{c}$ (shape: D)

Operations:
Alignment: $e_i = h \cdot x_i$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $\mathbf{c} = \sum_i a_i x_i$

Change $f_{\text{att}}(\cdot)$ to a dot product, this actually can work well in practice, but a simple dot product can have some issues...

Inputs:
Input vectors: $\mathbf{x}$ (shape: N x D)
Query: $\mathbf{h}$ (shape: D)

General attention layer



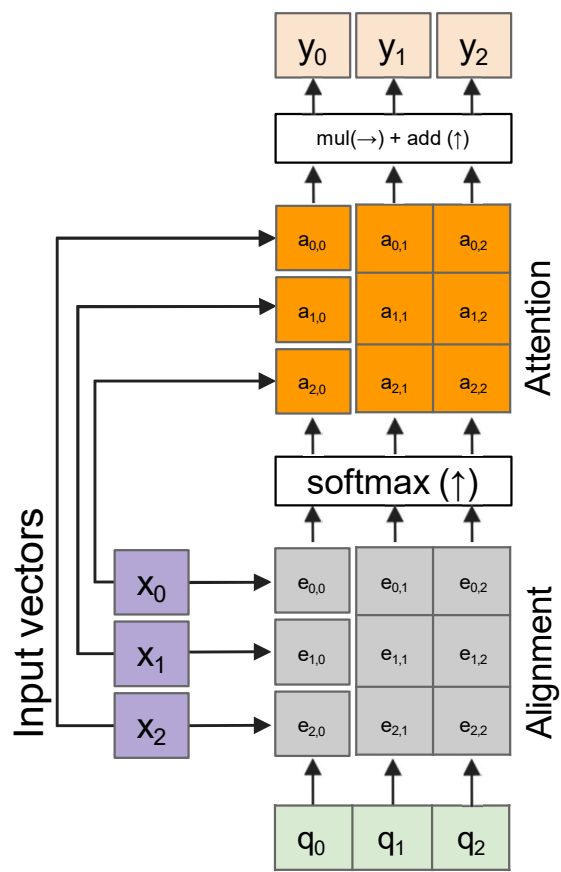
Outputs:
context vector: **c** (shape: D)

Operations:
Alignment: $e_i = h \cdot x_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $\mathbf{c} = \sum_i a_i x_i$

Inputs:
Input vectors: **x** (shape: N x D)
Query: **h** (shape: D)

- Change $f_{\text{att}}(.)$ to a **scaled** simple dot product
- Larger dimensions means more terms in the dot product sum.
 - So, the variance of the logits is higher. Large magnitude vectors will produce much higher logits.
 - So, the post-softmax distribution has lower-entropy, assuming logits are IID.
 - Ultimately, these large magnitude vectors will cause softmax to peak and assign very little weight to all others
 - Divide by $\sqrt{D}$ to reduce effect of large magnitude vectors
 - Similar to Xavier and Kaiming Initialization!

General attention layer



Outputs:
context vectors: $\mathbf{y}$ (shape: D)

Operations:
Alignment: $e_{i,j} = q_j \cdot x_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} x_i$

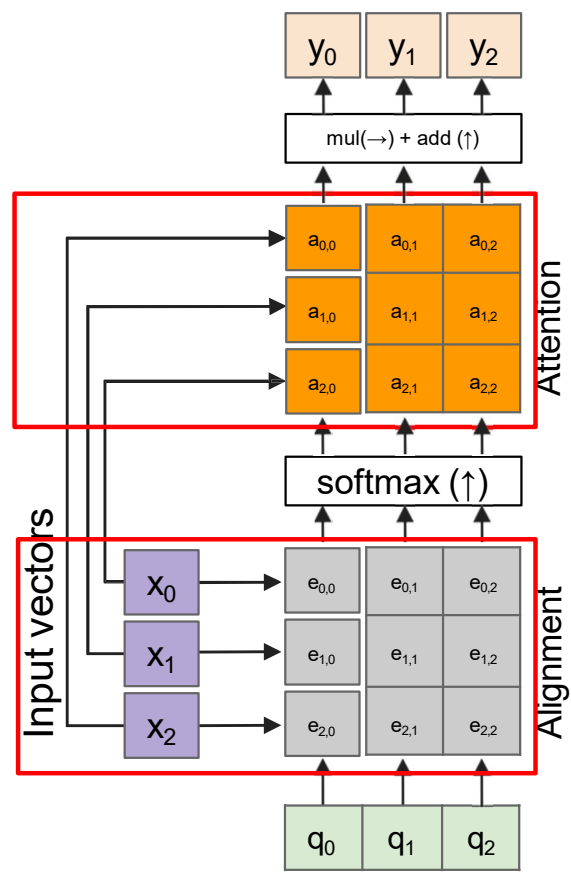
Inputs:
Input vectors: $\mathbf{x}$ (shape: N x D)
Queries: $\mathbf{q}$ (shape: M x D)

Multiple query vectors
- each query creates a new, corresponding output context vector

Allows us to compute multiple attention context vectors at once
Will go into more details in future slides, but this allows us to compute context vectors for multiple timesteps in parallel

Multiple query vectors

General attention layer



Outputs:
context vectors: y (shape: D)

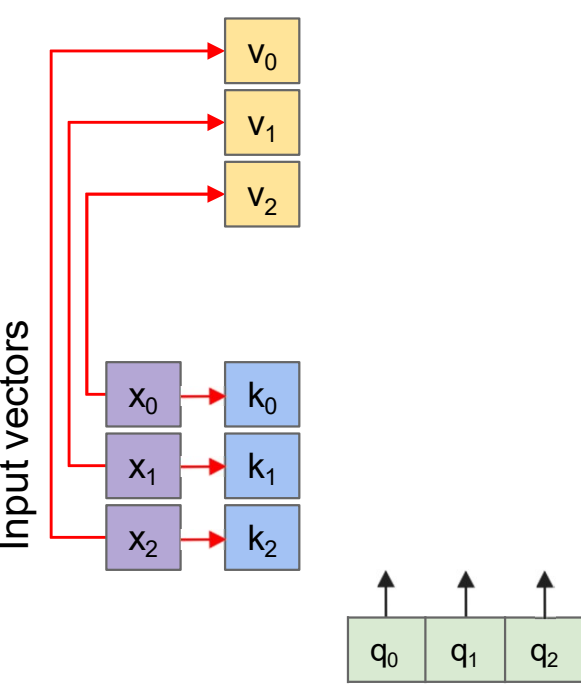
Operations:
Alignment: $e_{i,j} = q_j \cdot x_i / \sqrt{D}$
Attention: $a = \text{softmax}(e)$
Output: $y_j = \sum_i a_{i,j} x_i$

Inputs:
Input vectors: x (shape: N x D)
Queries: q (shape: M x D)

Notice that the input vectors are used for both the alignment as well as the attention calculations.

- We can add more expressivity to the layer by adding a different FC layer before each of the two steps.

General attention layer



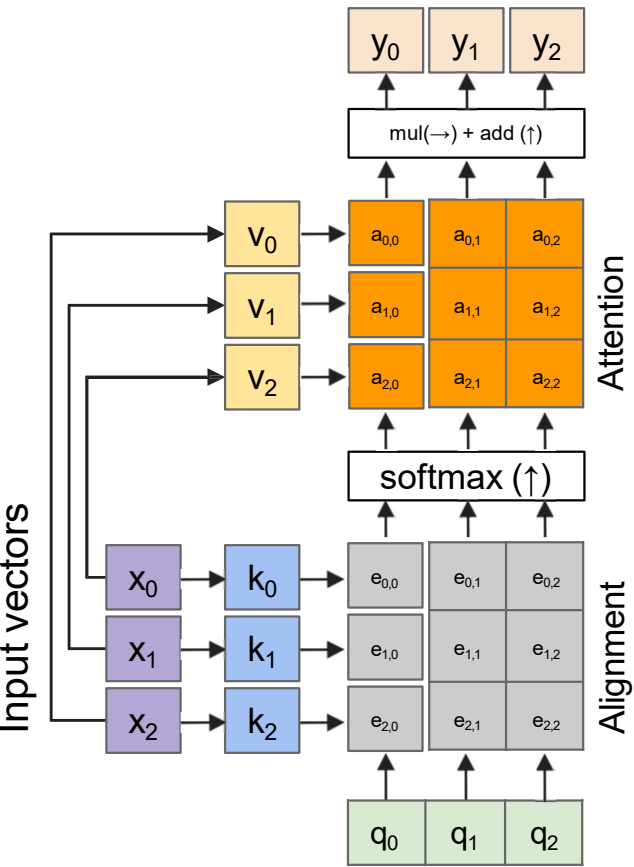
Operations:
Key vectors: $\mathbf{k} = \mathbf{x}\mathbf{W}_k$
Value vectors: $\mathbf{v} = \mathbf{x}\mathbf{W}_v$

Inputs:
Input vectors: $\mathbf{x}$ (shape: $N \times D$)
Queries: $\mathbf{q}$ (shape: $M \times D_k$)

Notice that the input vectors are used for both the alignment as well as the attention calculations.

- We can add more expressivity to the layer by adding a different FC layer before each of the two steps.

General attention layer



Outputs:
context vectors: $\mathbf{y}$ (shape: D_v)

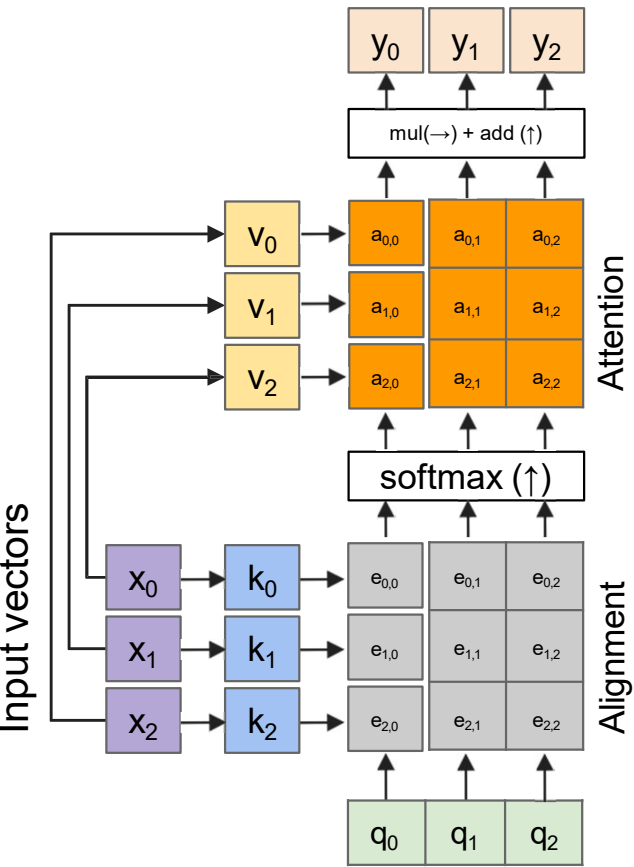
The input and output dimensions can now change depending on the key and value FC layers

Operations:
Key vectors: $\mathbf{k} = \mathbf{x}\mathbf{W}_k$
Value vectors: $\mathbf{v} = \mathbf{x}\mathbf{W}_v$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} v_i$

Since the alignment scores are just scalars, the value vectors can be any dimension we want

Inputs:
Input vectors: $\mathbf{x}$ (shape: $N \times D$)
Queries: $\mathbf{q}$ (shape: $M \times D_k$)

General attention layer



Outputs:
context vectors: y (shape: D_v)

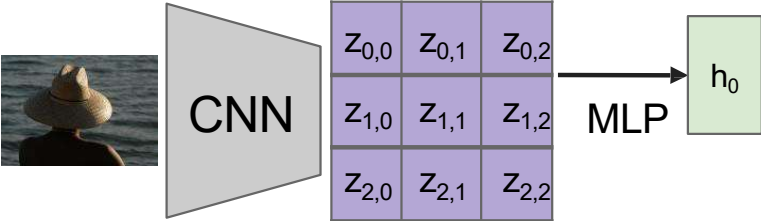
Operations:
Key vectors: $k = xW_k$
Value vectors: $v = xW_v$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $a = \text{softmax}(e)$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: x (shape: $N \times D$)
Queries: q (shape: $M \times D_k$)

This is a working example of how we could use an attention layer + CNN encoder for image captioning

Recall that the query vector was a function of the input vectors

Encoder: $h_0 = f_w(z)$
where z is spatial CNN features
 $f_w(\cdot)$ is an MLP

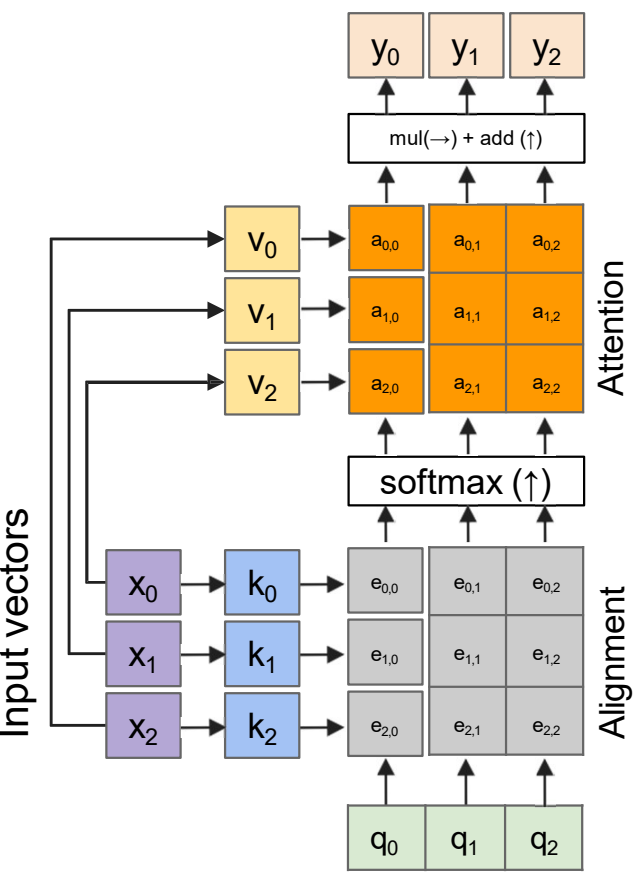


We used h_0 as q_0 previously

2. From Attention Layers to Transformers

Prof. Jianguo Zhang
SUSTech

Next: The Self-attention Layer



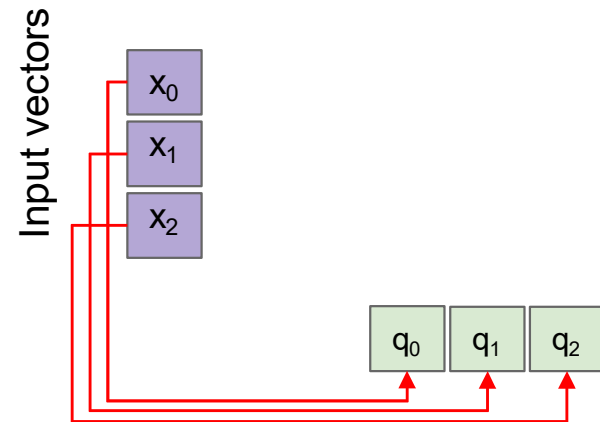
Outputs:
context vectors: $\mathbf{y}$ (shape: D_v)

Operations:
Key vectors: $\mathbf{k} = \mathbf{x}\mathbf{W}_k$
Value vectors: $\mathbf{v} = \mathbf{x}\mathbf{W}_v$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: $\mathbf{x}$ (shape: $N \times D$)
Queries: $\mathbf{q}$ (shape: $M \times D_k$)

Idea: leverages the strengths of attention layers without the need for separate query vectors.

Self attention layer



Operations:

Key vectors: $\mathbf{k} = \mathbf{x}\mathbf{W}_k$

Value vectors: $\mathbf{v} = \mathbf{x}\mathbf{W}_v$

Query vectors: $\mathbf{q} = \mathbf{x}\mathbf{W}_q$

Alignment: $e_{ij} = \mathbf{q}_j \cdot \mathbf{k}_i / \sqrt{D}$

Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$

Output: $y_j = \sum_i a_{i,j} v_i$

We can calculate the query vectors from the input vectors, therefore, defining a "self-attention" layer.

Instead, query vectors are calculated using a FC layer.

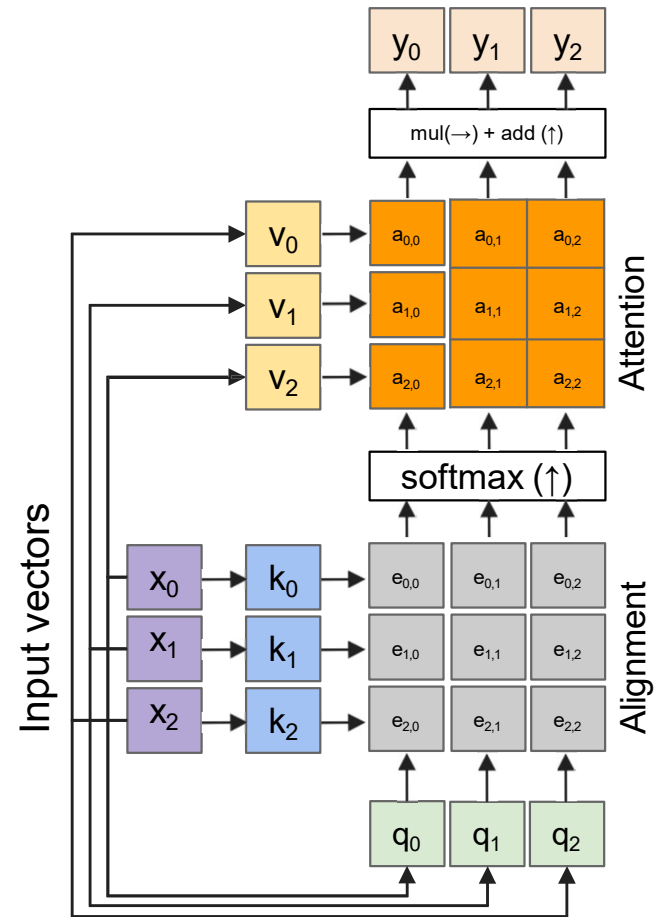
No input query vectors anymore

Inputs:

Input vectors: $\mathbf{x}$ (shape: $N \times D$)

Queries: $\mathbf{q}$ (shape: $M \times D_k$)

Self attention layer

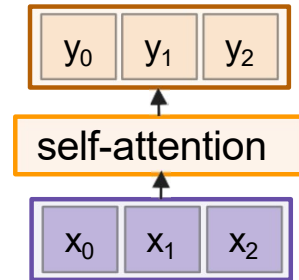
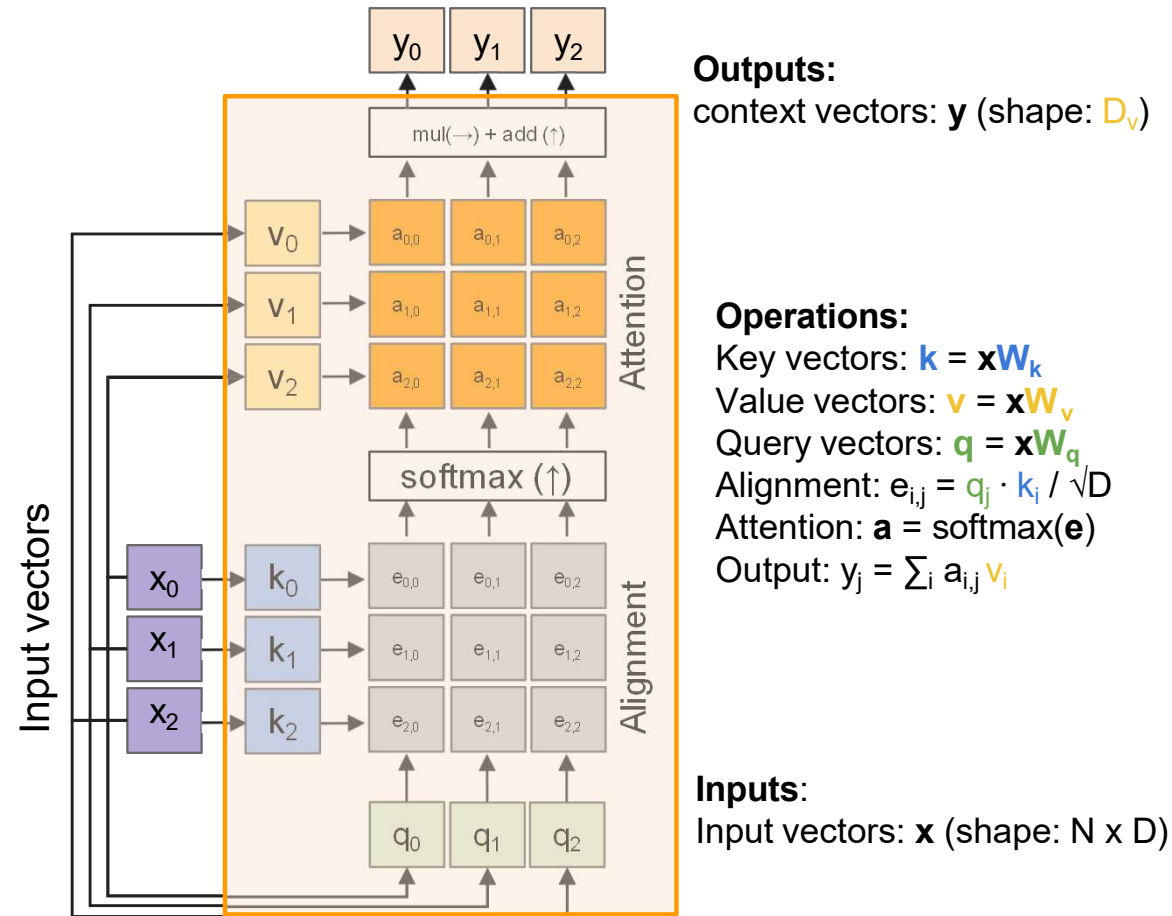


Outputs:
context vectors: $\mathbf{y}$ (shape: D_v)

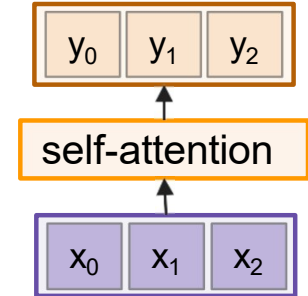
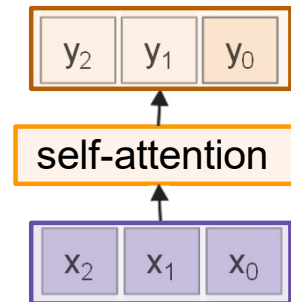
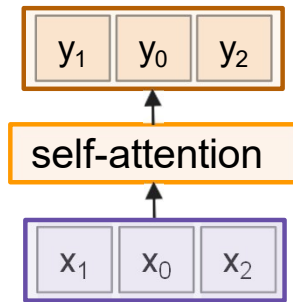
Operations:
Key vectors: $\mathbf{k} = \mathbf{x}\mathbf{W}_k$
Value vectors: $\mathbf{v} = \mathbf{x}\mathbf{W}_v$
Query vectors: $\mathbf{q} = \mathbf{x}\mathbf{W}_q$
Alignment: $e_{i,j} = \mathbf{q}_j \cdot \mathbf{k}_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: $\mathbf{x}$ (shape: $N \times D$)

Self attention layer - attends over sets of inputs



Self attention layer - attends over sets of inputs



Permutation equivariant

Self-attention layer doesn't care about the orders of the inputs!

Problem: How can we encode ordered sequences like language or spatially ordered image features?

Positional encoding

- The transformer so far ignores the sequential order information of the input sequence (i.e., the order of the words in a sentence)
- However: position and order of words are the essential parts of any language. They define grammar and semantics.
- The attention block in the transformer is invariant to the permutation of Key/Values for a give query. It can't tell the difference between the sequence of words of different ordering.
- *E.g., I eat apple versus Apple eat me.* Given the same query, the output of transformer will be the same.

Positional encoding

- Need to insert position information
- Positional embedding: a function mapping positions (of words) into vectors
- Ideally, the function should have the **following properties**:
 - It should output **a unique encoding** for each time-step (word's position in a sentence)
 - Distance between any two time-steps should be consistent across sentences with different lengths.
 - The model should generalize to longer sentences without any efforts. Its values should be bounded.
 - It must be deterministic.

Positional encoding

- To give transformer information about ordering of tokens, add function of position (based on sines and cosines) to every input

$$\vec{p}_t^{(i)} = f(t)^{(i)} := \begin{cases} \sin(\omega_k \cdot t), & \text{if } i = 2k \\ \cos(\omega_k \cdot t), & \text{if } i = 2k + 1 \end{cases}$$

where

$$\omega_k = \frac{1}{10000^{2k/d}}$$

- Note: this sin and cos functions satisfy all the requirements of the positional encoding function.

The intuition

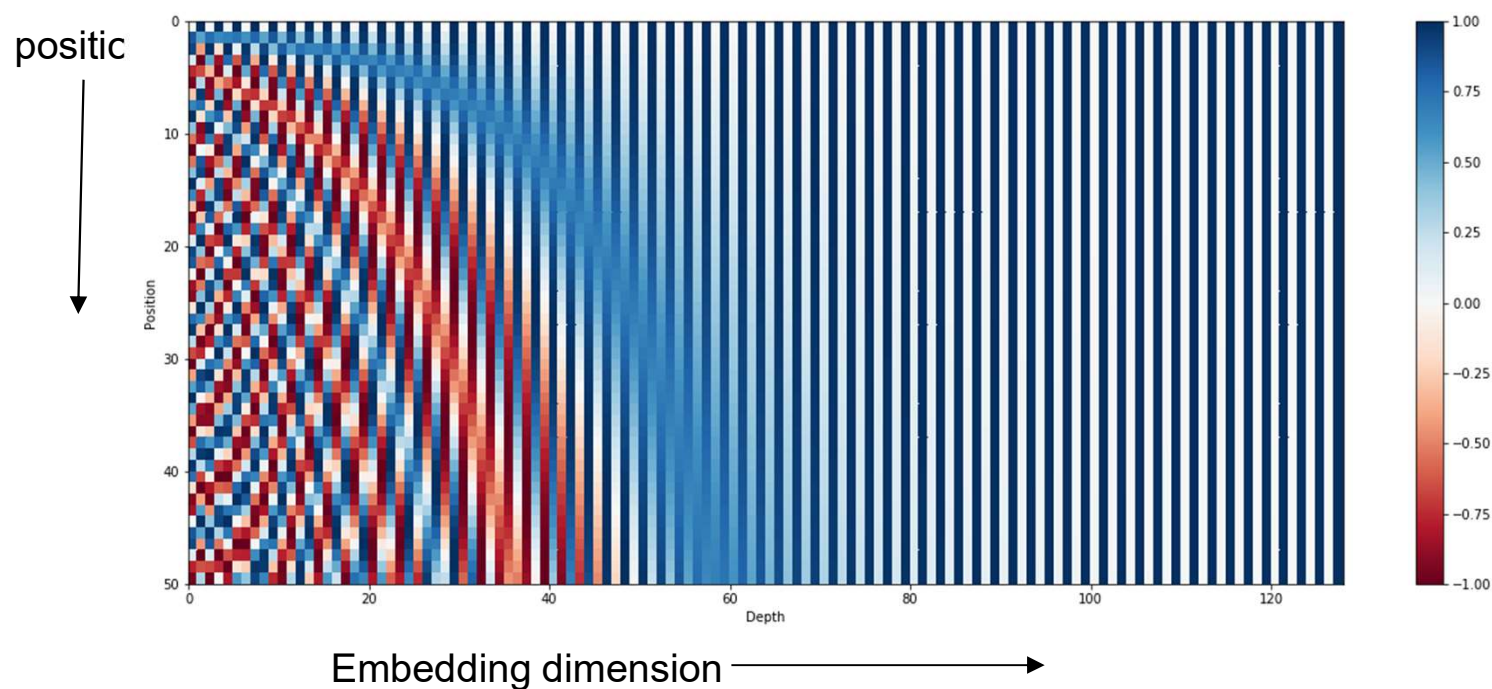
- You may wonder how this combination of sines and cosines could ever represent a position/order? It is actually quite simple,

0 :	0	0	0	0	8 :	1	0	0	0
1 :	0	0	0	1	9 :	1	0	0	1
2 :	0	0	1	0	10 :	1	0	1	0
3 :	0	0	1	1	11 :	1	0	1	1
4 :	0	1	0	0	12 :	1	1	0	0
5 :	0	1	0	1	13 :	1	1	0	1
6 :	0	1	1	0	14 :	1	1	1	0
7 :	0	1	1	1	15 :	1	1	1	1

**But using binary values
would be a waste of space
in the world of floats!!**

Positional encoding

- To give transformer information about ordering of tokens, add function of position (based on sines and cosines) to every input



The 128-dimensional positional encoding for a sentence with the maximum length of 50. Each row represents the embedding vector

[Image source](#)

Positional encoding

- **Observation:** Frequencies decreasing along depth for the give position. This is consistent with the trend in the positional binary coding.
- **Another property:** It allows the model to attend relative positions effortlessly. since for any fixed offset k , PE_{pos+k} can be represented as a linear function of PE_{pos} .
- Refer to <https://blog.timodenk.com/linear-relationships-in-the-transformers-positional-encoding/> for more details and proof.

Positional encoding

How to fuse the positional embedding to the token (e.g. words) embedding?

1. Add the positional embedding to the token embedding, requires the same number of the dimensions; but could save memories
2. Concatenate the positional embedding to the token embedding, dimensions can be different, however, require more memory spaces

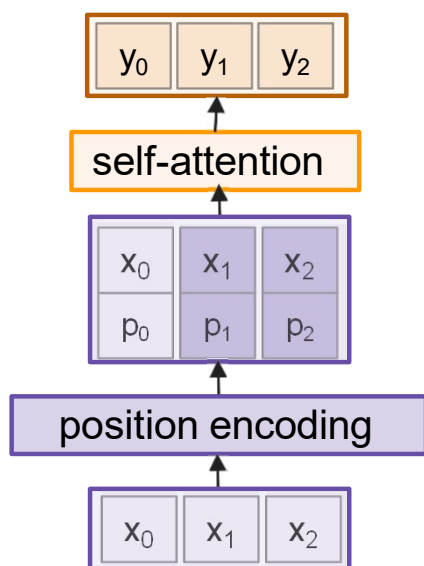
Positional embedding

Positional encoding is crucial for good performance of both text-transformer and vision-transformers.

1. So far we have introduced the **pre-defined position encoding** technique, which sometimes is called: **absolute positional** encoding. I.e., the mapping function (sin and cos) is fixed.
2. Positional embedding are also **learnable** (see the original Transformer paper) . Once learned, it is then fixed; **cause great difficulties in handling sequences longer than that seen in training.**
3. Other positional encodings:
 1. **Relative positional encodings** for text (by Shaw et al 2018) and for image (Bello et al 2019).
 2. **Complex-value encodings** (wang et al 2019). Rotary encodings in Roformer (Su et al 2021)
 3. **Conditional positional encoding** (Chu et al, ICLR2023).

For details see: Chu et al, Conditional Positional Encodings for Vision Transformers, ICLR 2023

Positional encoding



Concatenate or **add** special positional encoding p_j to each input vector x_j

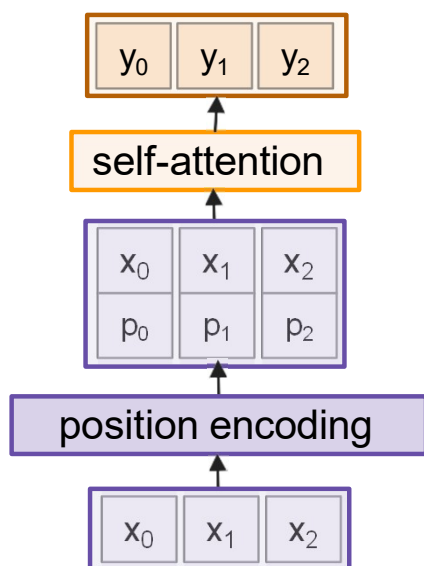
We use a function $pos: N \rightarrow R^d$ to process the position j of the vector into a d -dimensional vector

So, $p_j = pos(j)$

Possible desirable properties of $pos(.)$:

1. It should output a **unique** encoding for each time-step (word's position in a sentence)
2. **Distance** between any two time-steps should be consistent across sentences with different lengths.
3. Our model should generalize to **longer** sentences without any efforts. Its values should be bounded.
4. It must be **deterministic**.

Positional encoding



Concatenate special positional encoding p_j to each input vector x_j

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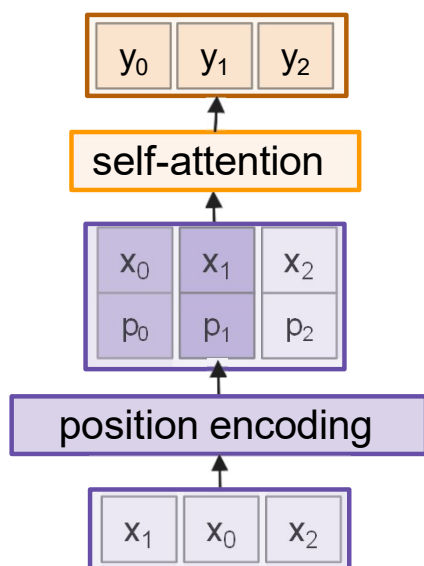
Options for $pos(.)$

1. Learn a lookup table:
 - Learn parameters to use for $pos(t)$ for $t \in [0, T)$
 - Lookup table contains $T \times d$ parameters.

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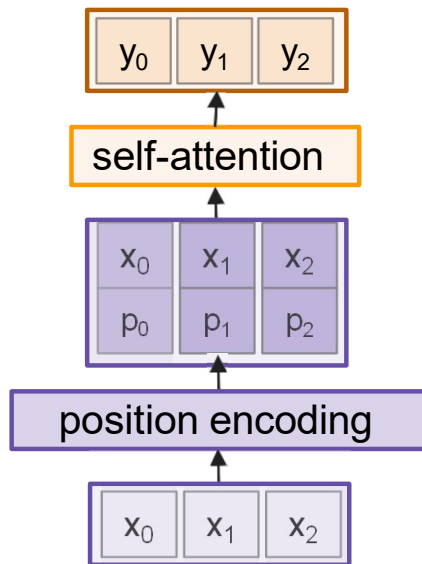
1. Learn a lookup table:
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2. Design a fixed function with the desired properties

$$p(t) = \begin{bmatrix} \sin(\omega_1 \cdot t) \\ \cos(\omega_1 \cdot t) \\ \\ \sin(\omega_2 \cdot t) \\ \cos(\omega_2 \cdot t) \\ \\ \vdots \\ \\ \sin(\omega_{d/2} \cdot t) \\ \cos(\omega_{d/2} \cdot t) \end{bmatrix}_d$$

$$\text{where } \omega_k = \frac{1}{10000^{2k/d}}$$

Vaswani et al, "Attention is all you need", NeurIPS 2017

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Intuition:

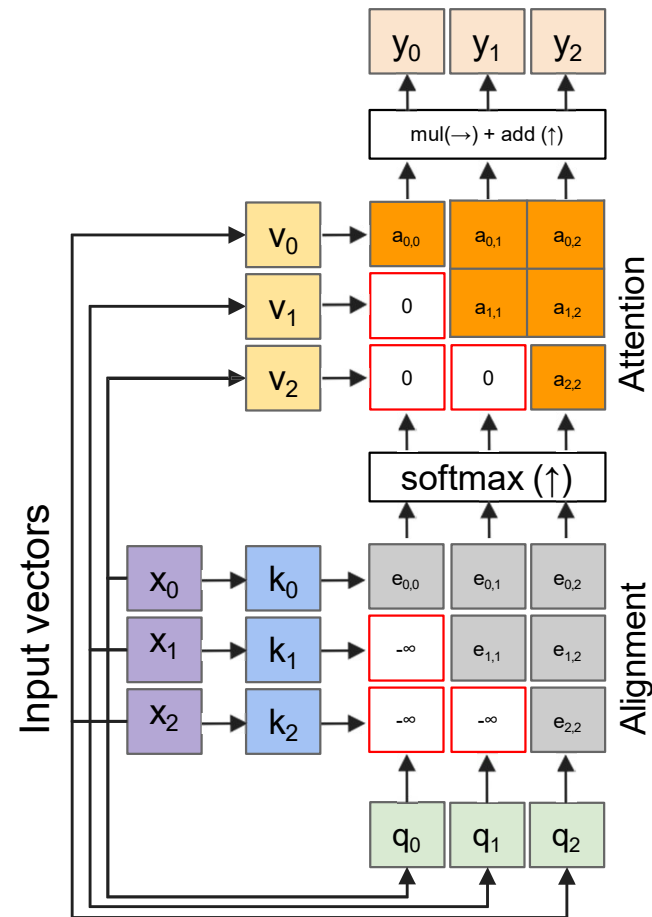
0 :	0	0	0	0
1 :	0	0	0	1
2 :	0	0	1	0
3 :	0	0	1	1
4 :	0	1	0	0
5 :	0	1	0	1
6 :	0	1	1	0
7 :	0	1	1	1
8 :	1	0	0	0
9 :	1	0	0	1
10 :	1	0	1	0
11 :	1	0	1	1
12 :	1	1	0	0
13 :	1	1	0	1
14 :	1	1	1	0
15 :	1	1	1	1

where $\omega_k = \frac{1}{10000^{2k/d}}$

[image source](#)

Vaswani et al, "Attention is all you need", NeurIPS 2017

Masked self-attention layer



Outputs:
context vectors: y (shape: D_v)

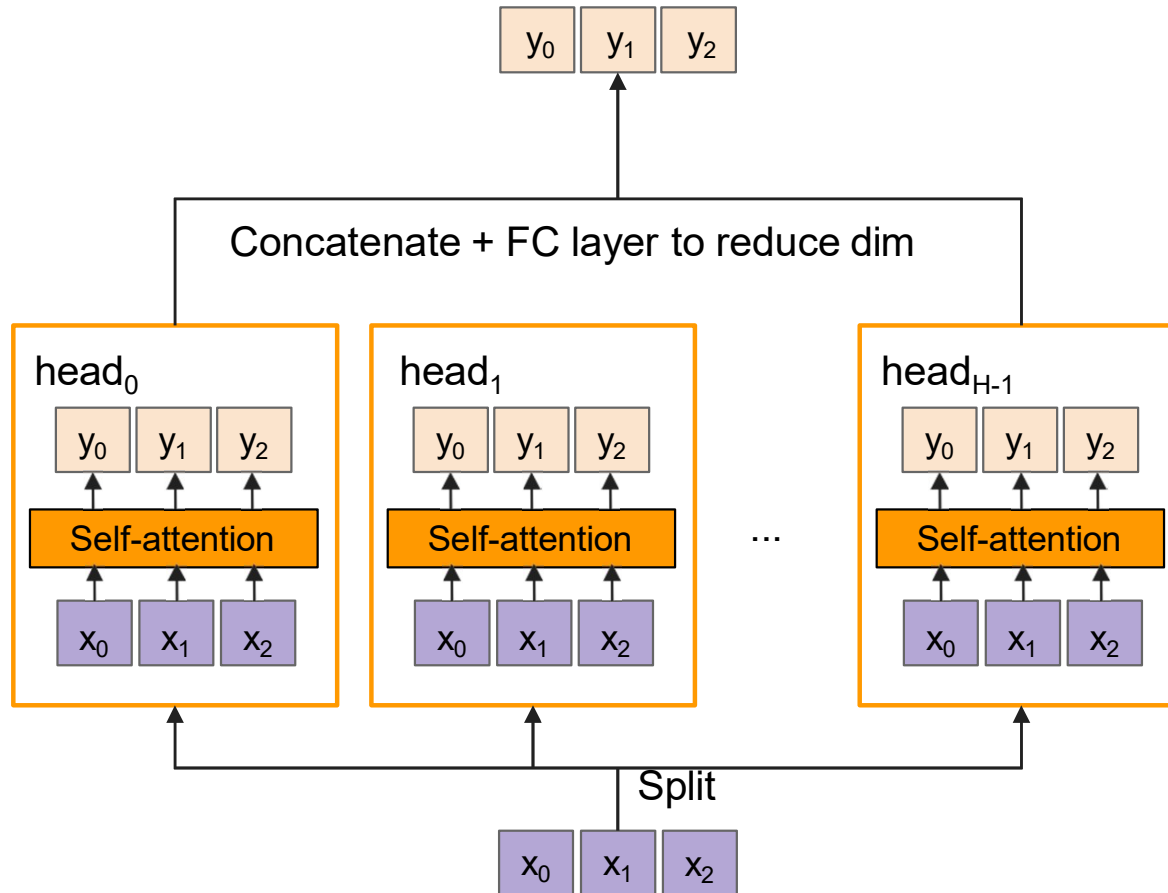
Operations:
Key vectors: $k = xW_k$
Value vectors: $v = xW_v$
Query vectors: $q = xW_q$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $a = \text{softmax}(e)$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: x (shape: $N \times D$)

- Allows us to parallelize attention across time
- Don't need to calculate the context vectors from the previous timestep first!
- Prevent vectors from looking at future vectors.
- Manually set alignment scores to $-\infty$ (-nan)

Multi-head self-attention layer

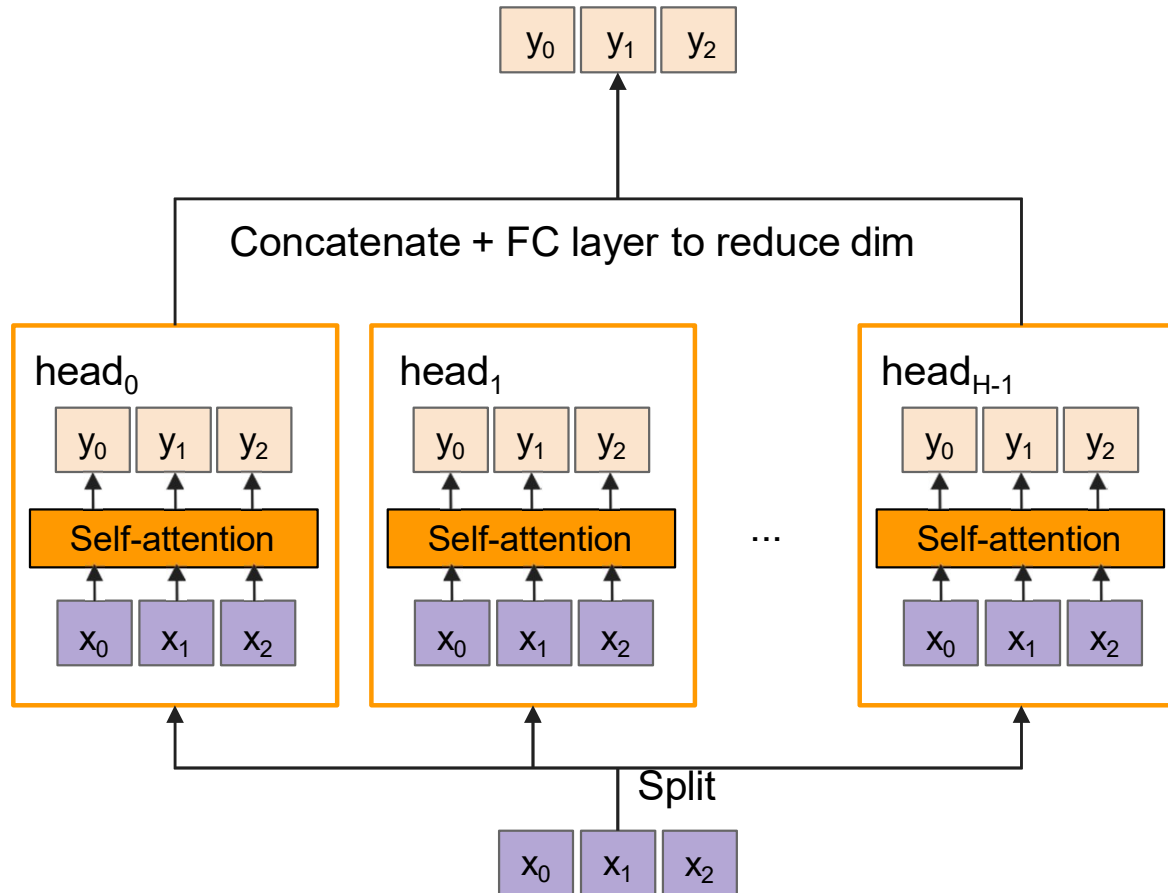
- Multiple self-attention “heads” in parallel



Q: Why do this?

Multi-head self-attention layer

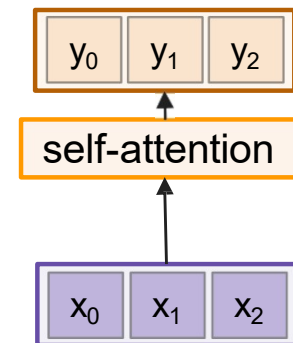
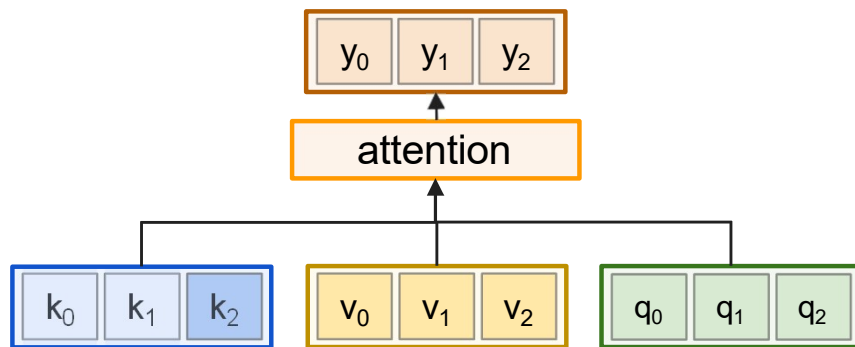
- Multiple self-attention “heads” in parallel



A: We may want to have multiple sets of queries/keys/values calculated in the layer. This is a similar idea to having multiple conv filters learned in a layer

General attention versus self-attention

Transformer models rely on many, stacked self-attention layers



Comparing RNNs to Transformer

RNNs

(+) LSTMs work reasonably well for long sequences. (-) Expects an ordered sequences of inputs

(-) Sequential computation: subsequent hidden states can only be computed after the previous ones are done.

Transformer:

(+) Good at long sequences. Each attention calculation looks at all inputs.

(+) Can operate over unordered sets or ordered sequences with positional encodings.

(+) Parallel computation: All alignment and attention scores for all inputs can be done in parallel. (-) Requires a lot of memory: $N \times M$ alignment and attention scalars need to be calculated and stored for a single self-attention head. (but GPUs are getting bigger and better)

Attention Is All You Need

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“ImageNet Moment for Natural Language Processing”

Pretraining:

Download a lot of text from the internet

Train a giant Transformer model for language modeling

Finetuning:

Fine-tune the Transformer on your own NLP task